Manuscript outline

Cell enlargement, polarity competition and migration persistence

Polarity replacement, spatial response maps and noise-driven heading memory

23 October 2026 · Review candidate; targeted mode campaign active

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes how established polarity is organized and replaced; these changes may provide a physical basis for greater migration persistence. The theoretical contribution tests size under unchanged local kinetic parameters. Length differs between model cells to represent WGD-associated enlargement, rather than elongation during a turn. Existing migration observations supply empirical motivation; new paired WGD motility results retain a substantial, explicitly reserved contribution.

Figure 2 separates size-dependent initiation and cue-width responses from temporal class signatures: initiation-limited resetting, saturating activation with visible gap retention, and pulse-2 progress after loss of visible activity. The full Holmes width series shows finite replacement at all tested lengths for sufficiently broad cues; narrow-cue censoring cannot establish a growing penalty.

Weak standard-pulse susceptibility changes with size in a model-dependent manner; its spatial map is validated at edge, active-interior and basal-arc positions (maximum error 1.69% of map amplitude). The covariance-weighted map predicts domain drift without fitting stochastic data. Wave-pinning f=0.001 C60/C20 drift remains 0.490 predicted and 0.473 measured; Goryachev remains quantitatively inconsistent. The weak-input alpha bridge retains unavailable observed exponents explicitly. Linear range at the right edge (first tested >10% gain departure, with last linear amplitude one doubling below): Wave-pinning: first >10% departures 0.02048 at C=20 and 0.01024 at C=60, size fold 0.5; last tested linear amplitudes 0.01024 and 0.00512; Holmes: first >10% departures 0.0512 at C=20 and 0.0512 at C=60, size fold 1; last tested linear amplitudes 0.0256 and 0.0256; Spring: first >10% departures 0.1024 at C=1 and 0.1024 at C=3, size fold 1; last tested linear amplitudes 0.0512 and 0.0512; Goryachev: first >10% departures 0.32768 at C=20 and 1.31072 at C=60, size fold 4; last tested linear amplitudes 0.16384 and 0.65536.

Manuscript outline · continued

Larger rings preserve heading longer while their local size exponent weakens as input strengthens. Completed region diagnostics show all-input closure bounds and remainder curves. The independent modes-only campaign is active; this review candidate claims no completed mode-collapse result.

The prospective pilot table fixes model-linked cellular observables before the SUM-159 data arrive. Directions are conditional hypotheses; unavailable model measurements remain unavailable. Activity heading, observed migration summaries and conditional OU velocity-memory estimates remain separate evidence.

This is a working manuscript outline. Figures 1–2 retain their numerical evidence with separate panels for distinct quantities. The published migration evidence in Figure 5, reserved Figure 6 and prospective pilot appendix retain their preceding content. Activity polarity, modelled replacement, observed migration and conditional OU velocity memory remain distinct readouts.

Proposed main figures

- Figure 1. Enlargement changes the cue duration required for lasting polarity replacement
- Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures
- Figure 3. Size-dependent susceptibility links deterministic perturbations to weak-input domain drift
- Figure 4. Enlargement preserves heading as noisy polarity changes regime.
- Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation
- Figure 6. RESERVED: new WGD motility-assay results

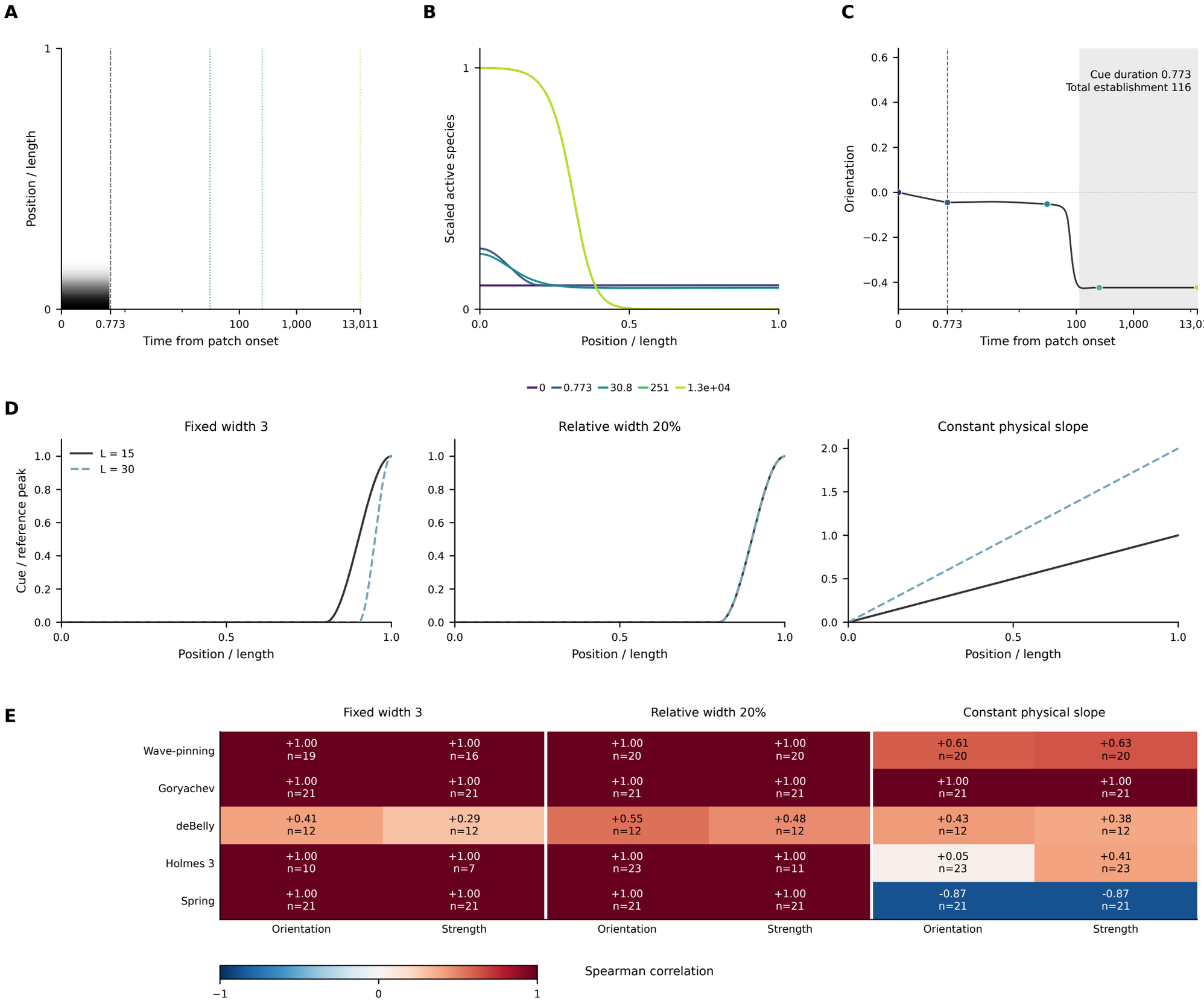

A
Position / length
Time from patch onset
B
Scaled active species
Position / length
0
0.773
30.8
251
1.3e+04
C
Cue duration 0.773
Total establishment 116
Orientation
Time from patch onset
D
Fixed width 3
Relative width 20%
Constant physical slope
L = 15
L = 30
Cue / reference peak
Position / length
E
Fixed width 3
Relative width 20%
Constant physical slope
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
+1.00 n=19
+1.00 n=16
+1.00 n=20
+1.00 n=20
+0.61 n=20
+0.63 n=20
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+0.41 n=12
+0.29 n=12
+0.55 n=12
+0.48 n=12
+0.43 n=12
+0.38 n=12
+1.00 n=10
+1.00 n=7
+1.00 n=23
+1.00 n=11
+0.05 n=23
+0.41 n=23
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
-0.87 n=21
-0.87 n=21
Orientation
Strength
Orientation
Strength
Orientation
Strength
Spearman correlation

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement

Comparative theoretical foundation: five established models, three finite challenges, and two distinct scalar replacement gates.

Detailed legend

A, Recorded Wave-pinning preparation cue at length 15: a finite left patch, withdrawn at 0.773 native time units. White to black is zero to peak input; coloured vertical lines mark the profile samples in B.

B, Native active profiles from the same preparation, scaled using the common minimum and span of the displayed snapshots. Colours identify recorded times, with no new simulation.

C, Orientation from the same preparation. Grey marks basal verification after establishment arrival at about 116; the cue has already ended. The matched time scale in A and C separates input duration from autonomous development.

D, Three rightward finite challenges restart the identical qualifying nominal-amplitude, relative-20% basal checkpoint at each model and length: fixed physical support 3 (Spring 0.2 native length units), relative support 20%, and a linear ramp with constant physical slope. Patch peaks remain fixed within each model; a fixed-slope ramp has a larger peak in a longer domain. Goryachev patches use the independently selected, length-independent multiplier K=64 (peak 6.4), which makes this a response-permissive comparison rather than a calibrated equal biochemical dose across models.

E, Spearman rank correlation between length and the shortest tested passing pulse duration, separately for orientation replacement and strength replacement, across Wave-pinning, Goryachev, corrected deBelly, Holmes 3 and Spring. Each cell reports the correlation and included successful-length count. All audited scalar successes enter, including those failing the separate native-field tail criterion. At least four distinct successful lengths and nonconstant durations are required; censored and preparation-ineligible cases receive no invented time. The companion duration curves retain all lengths and field-status markers.

After each positive pulse, input returns to the native basal value for 10,000 native time units. Orientation requires all 201 final-500 samples to have scalar orientation above 0.05 and contrast at least 0.05, with orientation range below 0.005. Strength additionally requires every tail orientation to be within 1% of the absolute initial leftward scalar orientation; it does not compare individual pole heights or amounts. The separate native-field tail flag requires pointwise range and projected 500-unit drift within 0.1% of frozen field scales. The reported duration excludes preparation and recovery. It is the shortest passing tested duration, with sampled fail/pass boundaries refined to 1%; narrower unobserved intervals are not excluded.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

deBelly uses independently commanded corrected endpoints, rather than the deprecated in-loop centered extension. Spring length and time retain native units and are not silently converted to micrometres or seconds. Model-specific axes and stimulus definitions prevent interpreting cross-model amplitudes or clocks as biologically calibrated equivalents.

Narrative points and conclusions

- Fixed-width and relative-width patches show perfectly increasing rank order among observed successes in Wave-pinning, Goryachev, Holmes 3 and Spring; enlargement often increases the finite stimulation required for a retained response.
- The response depends on the cue geometry and model. Spring has a negative constant-slope association (Spearman correlation -0.874 for both gates), while Holmes orientation under the constant-slope cue is nearly unassociated with length (0.047).
- Corrected deBelly curves are nonmonotone despite moderate positive rank correlations. The full duration curves and eligibility ranges are essential to the result.
- These are deterministic polarity-replacement experiments. They motivate, but do not measure, spontaneous polarity memory or migration persistence.

Proposed measurements and experimental follow-up

Mechanism-linked cellular follow-up would match local input support, amplitude and duration, track the original and challenged activity regions through stimulation and withdrawal, and distinguish orientation replacement from recovered activity strength. This validation is proposed and has no experimental results in this outline.

Scope, limitations and pending validation

Correlations are conditional on observed scalar success, rather than an unbiased summary of all possible sizes or a universal scaling law. Ineligible starts are distinct from responses unobserved by the finite horizon.

A retained scalar response can coexist with a changing native field. Only 13 of the 72 corrected deBelly scalar successes pass the separate native-field tail flag. Resolution checks from the earlier centered-deBelly campaign cannot qualify the corrected replacement thresholds.

Sources and reproducibility

Three-test protocol and Spearman rule

Correlation table

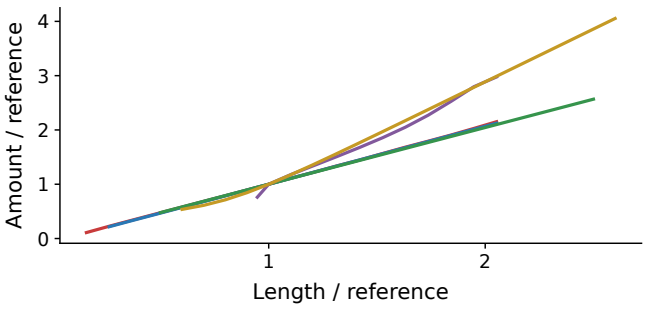
Corrected endpoint method

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

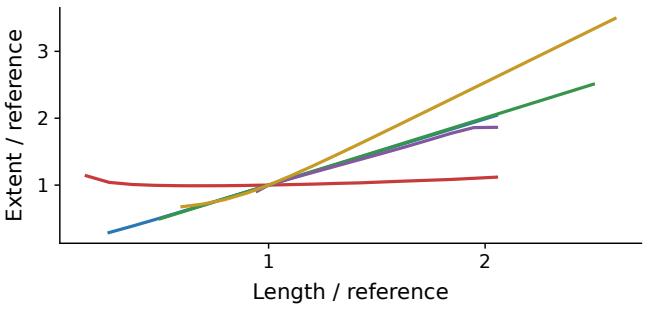
Current duration curves

Goryachev Wave-pinning Holmes deBelly Spring

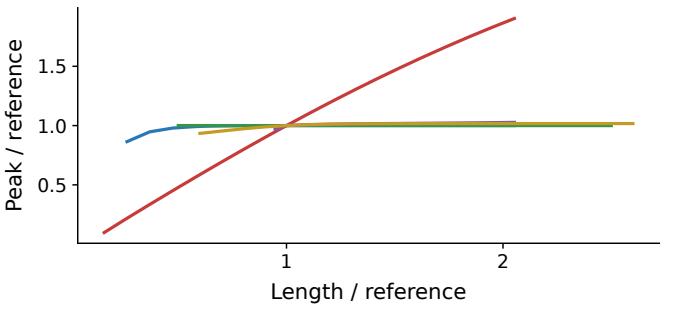
A



B



C

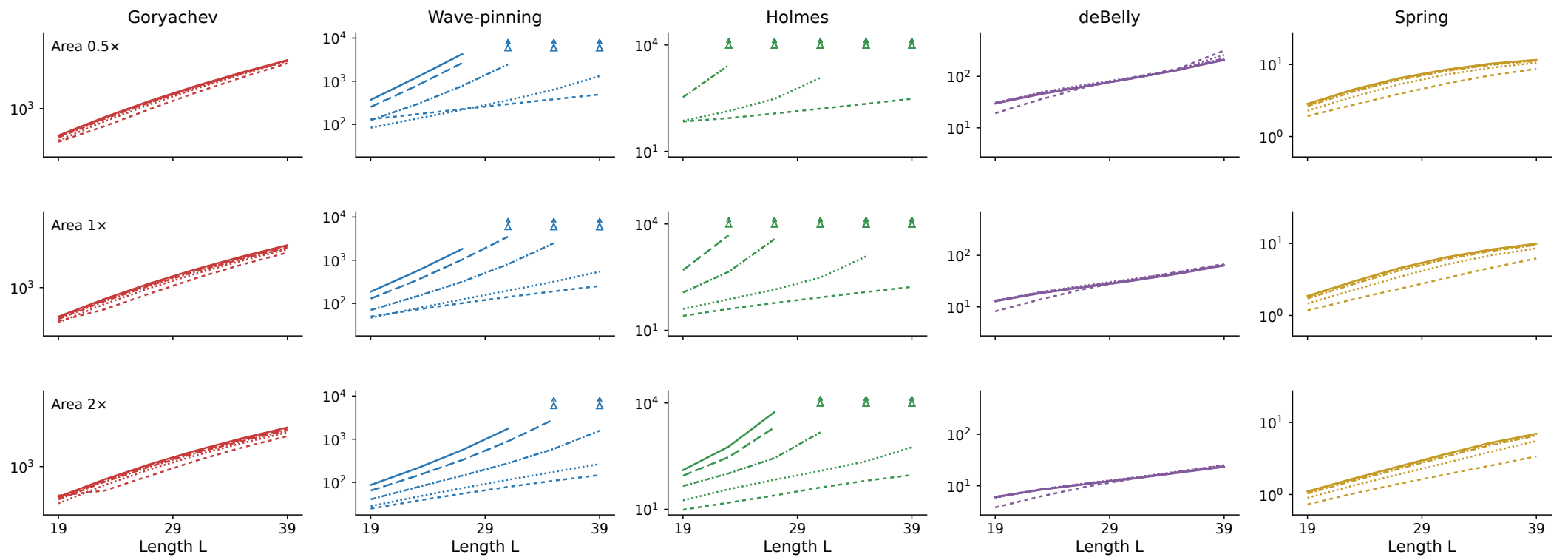


D

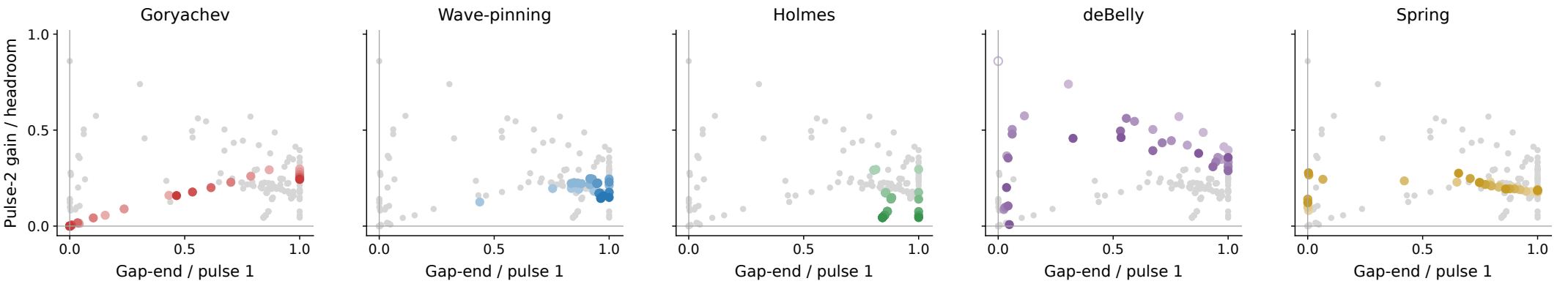


E

w = 0.95 w = 1.9 w = 3.8 w = 7.6 w = 15.2

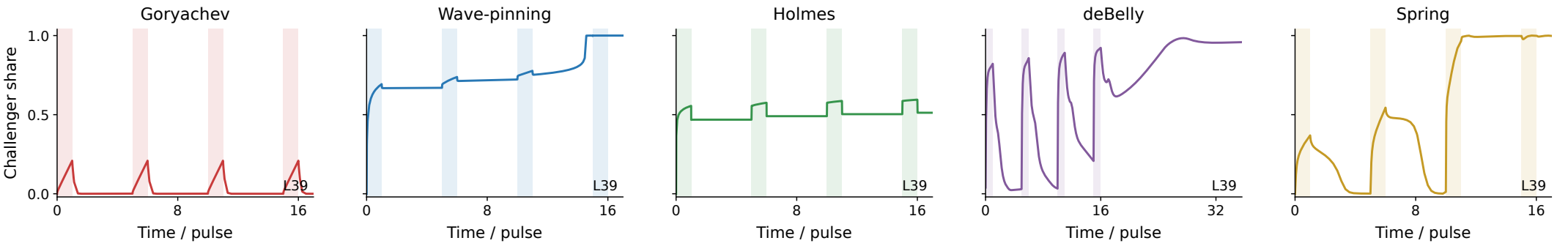


F



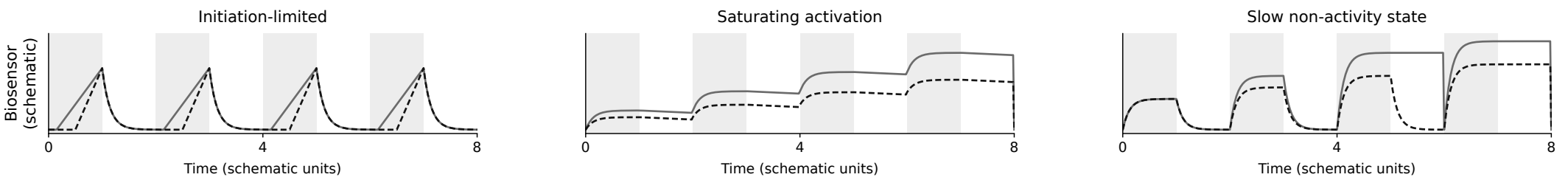
Open circles: below activity gate at x = 0. Size increases from light (L19) to dark (L39).

G



H

WGD− WGD+



I

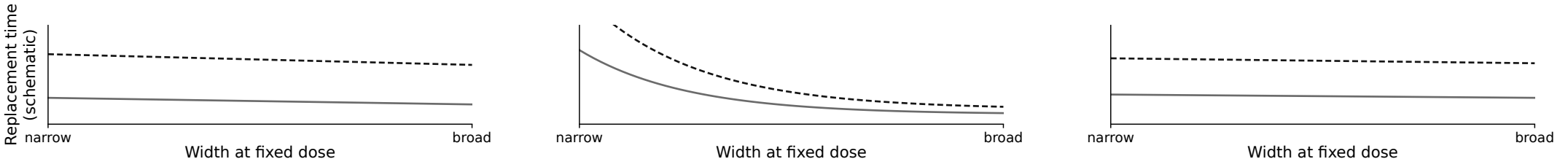


Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures

Separate size-dependent initiation and cue-width responses from observable pulse-memory classes.

Detailed legend

Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.

A, Incumbent excess amount normalized to the model reference, across the same 97 retained states.

B, Incumbent extent normalized to the model reference, for those same states.

C, Incumbent peak normalized to the model reference, for those same states.

D, Initiation is the time from cue onset until maximum in-cue activity minus the profile minimum exceeds 10% of the model's L19 incumbent excess peak. Area 1, width 3.8; times are normalized to L19. All six deBelly upper bounds are 0.5 native units (downward arrows); first-update sampling leaves its size dependence unresolved.

E, Replacement time (first eligible challenger share ≥ 0.75) versus length, with a line per width and rows for areas 0.5, 1 and 2. Upward triangles/arrows are lower bounds: the cue never replaced within the horizon (6,000 Wave-pinning; 10,000 Goryachev/Holmes; 2,000 deBelly; 100 Spring, native units). Broad cues rescue Holmes across all lengths; narrow-cue censoring does not establish a growing penalty. Spring native widths equal the displayed widths divided by 38.

F, Gap-1 activity kept versus pulse-2 gain as a fraction of headroom (Eq. 1). Four equal pulses, total ON 1.5 times the model-length commitment calibration; gaps 0–16 pulse durations. Holmes L31–39 uses the L27 calibration. Other models are grey; lightness indicates L. Open circles at x = 0 code activity below the existing visibility gate. At the left wall, deBelly/Spring gain on pulse 2 despite no visible activity; Goryachev never does. The deBelly L19/gap16 first-gap endpoint was checked against native fields.

G, L39 gap4 challenger-share histories, labelled kept / lost / lost then commits; shading marks cue ON. deBelly includes full recovery; others show one post-train pulse duration. Continuous raw shares include below-gate values. deBelly's later timing and fields remain numerically sensitive (supplement).

H, Prospective biosensor histories for the same pulse train in WGD− and enlarged WGD+ cells. The three facets show delayed rises and resets; saturation with gap retention; and gap collapse followed by commitment. All curves are dimensionless schematics, exactly as in the preceding edition.

I, Prospective equal-dose width responses for the same three classes: weak width rescue, strong width rescue, and weaker width rescue. Solid grey: WGD−; dashed black: WGD+.

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures · continued

These are unchanged schematics, not measurements. Model kinetics are unchanged.

Narrative points and conclusions

- Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.
- The full width series shows that Holmes can replace at every tested length when the cue is broad enough; narrow-cue censoring cannot establish a growing width-penalty ratio.
- Pulse-2 headroom gain separates low visible retention with residual progress from the initiation-limited reset response. Saturating activation retains visible activity through gaps.

Proposed measurements and experimental follow-up

Measure first in-cue biosensor rise, then vary width at fixed integrated dose.

Measure pulse-1, gap-1 and pulse-2 challenger shares under the same pulse train, followed by eventual commitment.

Scope, limitations and pending validation

Commanded input areas and native clocks are not calibrated across models.

deBelly initiation is unresolved within the first native update; later-gap timing and endpoint fields remain numerically sensitive.

F codes known below-gate activity at x=0; the exact raw ratio remains in the plotted-value CSV.

Holmes large-length trains have no target-length commitment calibration; visible retention differs from eventual replacement.

The three classes describe selected deterministic regimes; WGD biosensor predictions are prospective.

Model definitions and readouts

$$x = \frac{s_{\mathrm{gap}\,1}}{s_1}, \qquad y = \frac{s_2 - s_1}{1 - s_1} \tag{1}$$

s_1 and s_2 are challenger shares at pulse ends 1 and 2; s_gap1 is the first-gap-end share. An observed endpoint below the existing visibility gate is coded as zero visible share in F; its raw value remains in the CSV.

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures · continued

Sources and reproducibility

All 450 per-width replacement times

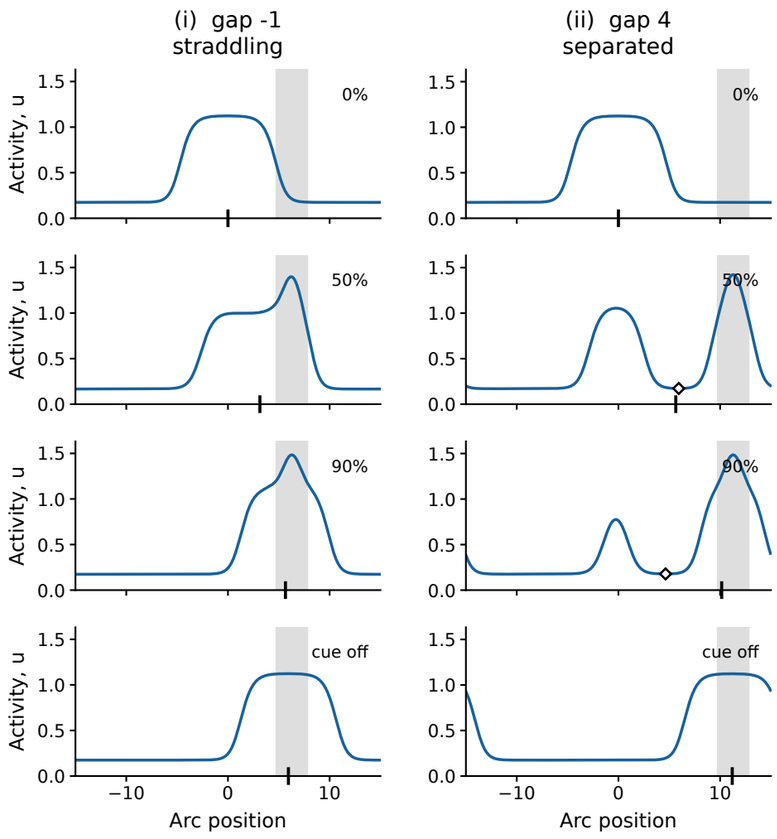
All 180 gap1/pulse2 coordinates and raw shares

deBelly L19/gap16 native-phase audit

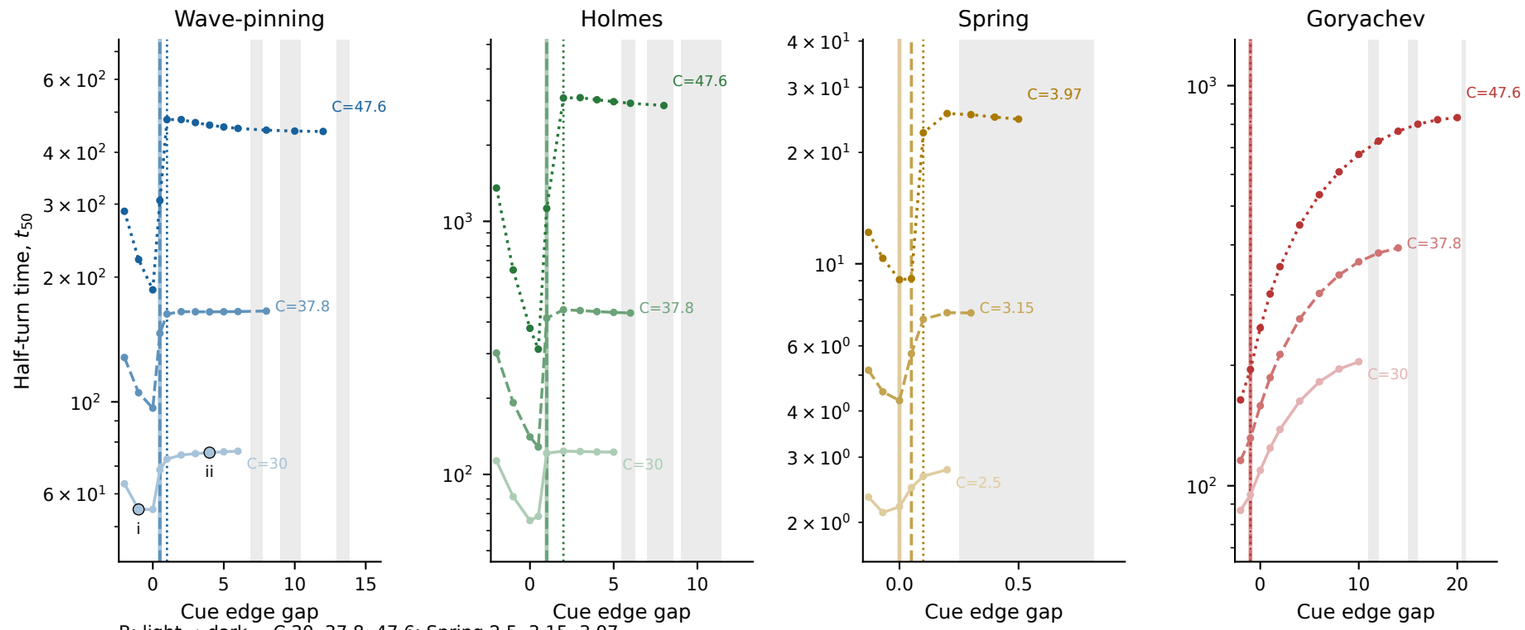
Preserved prior main panels and complete supplement

Grey: cue support; diamonds: saddle; | heading

A



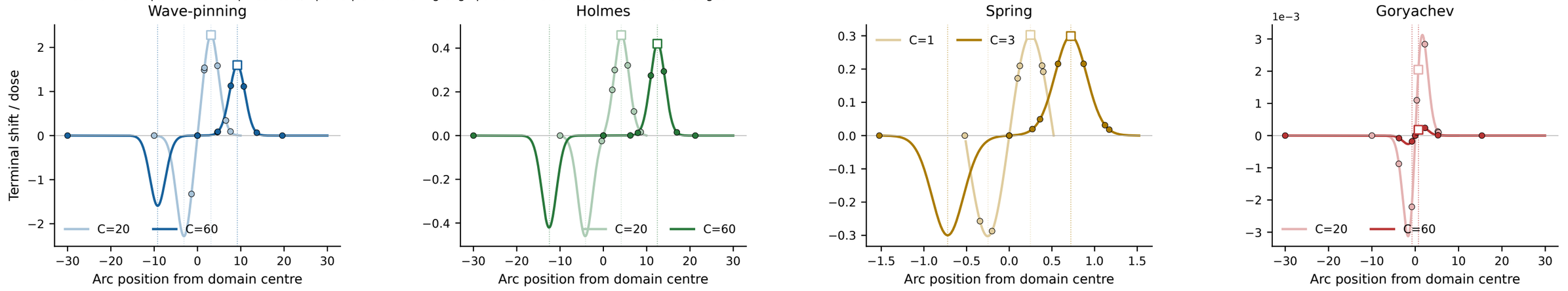
B



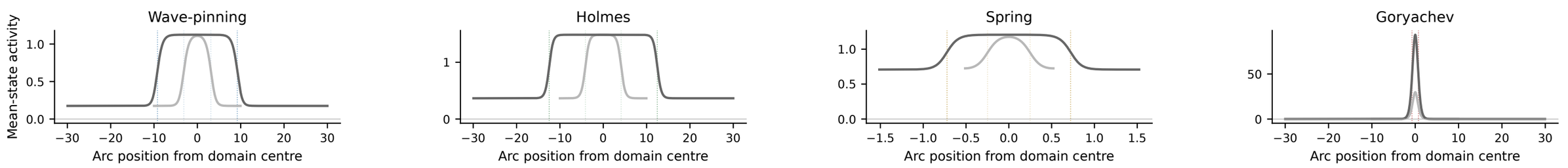
B: light → dark = C 30, 37.8, 47.6; Spring 2.5, 3.15, 3.97.
Open triangles: time lower bounds. Dotted lines: first saddle. Grey: heading undefined.

C

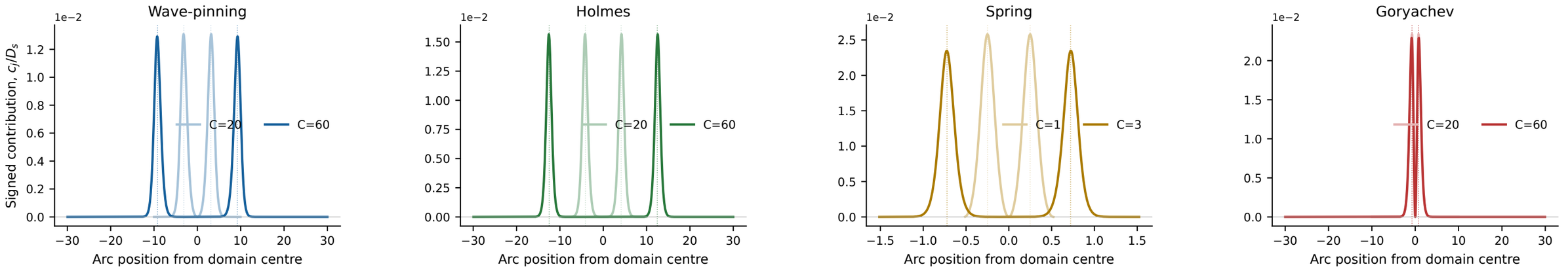
Lines: native map; circles: new pulse tests; open squares: existing edge pulse. Dotted verticals: half-contrast edges.



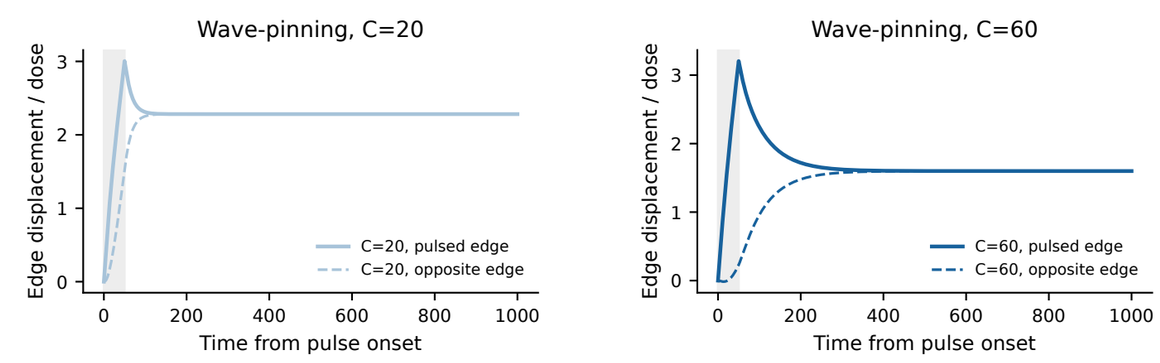
D



E



F

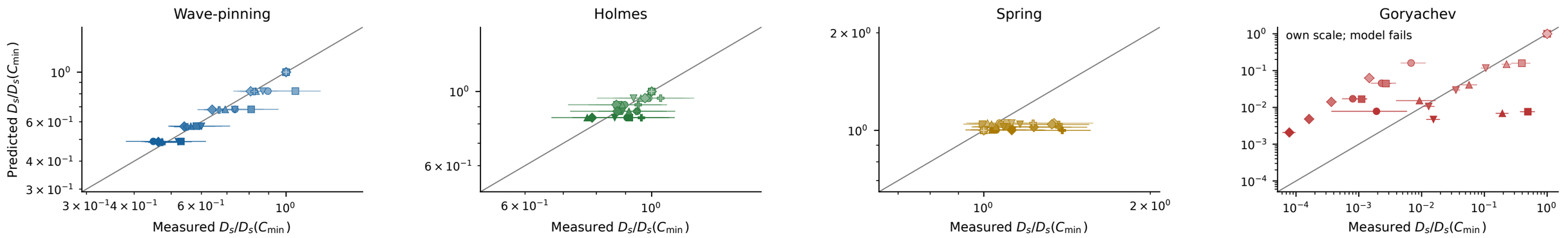


Permanent shift, both edges together

Terminal gain Cmax/Cmin:
Wave-pinning 0.701; Holmes 0.914
Spring 0.989; Goryachev 0.088

Weak input anywhere shifts the domain by g(x) × dose.
Noise is many such inputs; its covariance and g predict drift.
No drift data or model parameters are fitted.

G



H

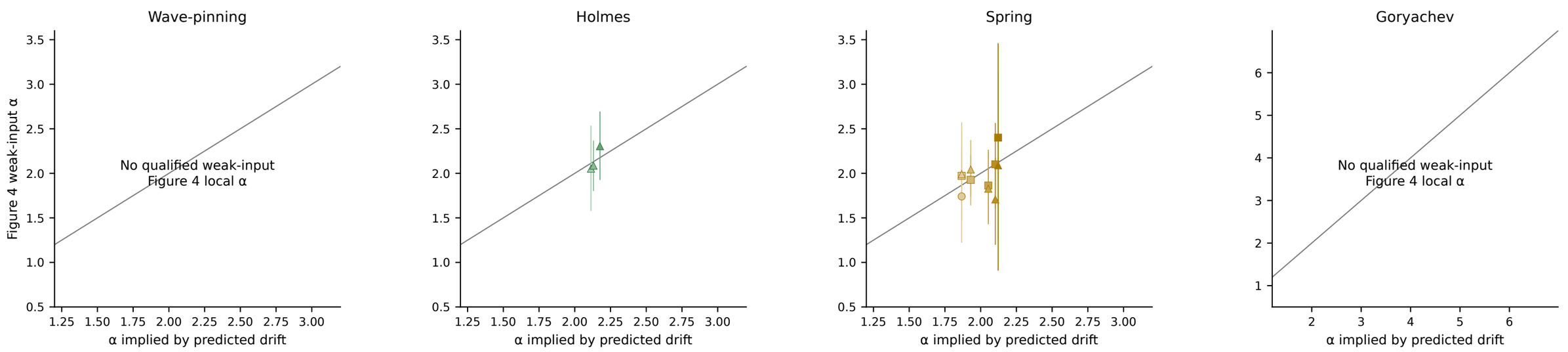


Figure 3. Size-dependent susceptibility links deterministic perturbations to weak-input domain drift

Spatial susceptibility and signed covariance contributions expose the deterministic input-to-drift bridge.

Detailed legend

A. Exact preceding Wave-pinning activity profiles, C=30, for a cue straddling the incumbent edge (i) or separated from it (ii). Rows are initial, half-turn, 90% turn and cue-free relaxation; the grey band is cue support, diamonds are saddles and the black tick is heading. All A data and timings are unchanged.

B. Exact preceding half-turn time against cue-edge gap for the three matched sizes, light to dark: C=30,37.8,47.6; Spring 2.5, 3.15, 3.97. Lines mark first saddle appearance; grey marks undefined heading at one or more sizes. Open upward triangles and arrows denote lower bounds. Continuous-cue data and native parameters are unchanged.

C. A weak input anywhere shifts the domain by g(x) times dose; noise is many such inputs; drift follows from g and the input covariance; nothing is fitted. Lines are the terminal displacement per dose predicted by the linearized native model for the standard Gaussian pulse centred at each arc position: SD 1.5, duration 50 (Spring 0.15, 5), at f=0.001. Light and dark are C20/C60 (Spring 1/3). New nominal-amplitude tests from settled states are filled circles; the existing edge pulse is an open square. Both subtract a matched unpulsed control. The pulse confirms g at the edge; the map elsewhere comes from the linearized native model and is tested at domain centre, halfway to the edge, edge±1/3 pulse SD, quarter-basal arc and antipode. Maximum absolute gain error normalized to map amplitude is 1.69%. Terminal edge gain large/small: Wave-pinning 0.701, Holmes 0.914, Spring 0.989, Goryachev 0.088.

D. Mean-state activity in light grey on its own axis, sharing the centred physical arc coordinate with C and E. Dotted verticals locate half-contrast edges. Spring positions are material fractions times the mean-state circumference.

E. Signed per-position contribution c_j/D_s at the same sizes and input, using raw native g and the unchanged noise covariance, with no measured-gain rescaling. Contributions sum to one; negative values are retained. Edge zones are within two input correlation lengths of the nearest edge; plateau interior and basal arc exclude these zones. Fractions of D_s, left/right edge, plateau interior, basal arc respectively: Wave-pinning C20: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000; Wave-pinning C60: left edge 0.499, right edge 0.499, plateau interior 0.001, basal arc 0.000; Holmes C20: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000; Holmes C60: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000; Spring C1: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000; Spring C3: left edge 0.498, right edge 0.498, plateau interior 0.002, basal arc 0.002; Goryachev C20: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000; Goryachev C60: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000.

Figure 3. Size-dependent susceptibility links deterministic perturbations to weak-input domain drift · continued

Raw-native and measurement-normalized totals are both exported.

F. Wave-pinning C20 and C60: pulsed falling edge (solid) and opposite rising edge (dashed) move together and retain a permanent offset after the grey pulse-on interval. Displacement is divided by pulse dose. Width and centroid histories, other models, every amplitude and duration are supplementary.

G. Predicted against measured physical domain diffusion after each is divided by its own fixed-smallest-size value at the same input. Only the shared-translation prediction appears here; the grey line is equality. The main prediction uses raw native g without measured-gain rescaling; the historical pulse-normalized variant is retained in the supplement. Shape encodes f=0.0001,0.0003,0.001,0.003,0.005,0.01 (circle,square,up/down triangle,diamond,plus); size retains the model colour ramp. Horizontal intervals are measured bootstrap 95% intervals. The ratio is approximately the squared edge-gain fold: Wave-pinning $0.70^2 \approx 0.49$; the full spatial covariance gives 0.490, versus 0.473 measured at f=0.001. Shape changes make the square relation approximate, not an exact identity. Absolute predictions and independent-edge candidate are supplementary. Goryachev stays on its own scale and fails quantitatively.

H. Local $\alpha = 2 - d\ln D_s / d\ln C$ implied by the predicted drift against Figure 4 weak-input local α (f≤0.001), with identical fixed-neighbour size fits, two neighbours at endpoints and three inside. Grey is equality. Vertical intervals are Figure 4 bootstrap 95% intervals. 14 of 60 requested pairs qualify; Wave-pinning and Goryachev have no qualified weak-input observed α under the fixed shared-lag rule. No missing estimate is replaced. Marker shapes distinguish the three weak inputs as in G.

Narrative points and conclusions

- Weak standard-pulse susceptibility changes with size in a model-dependent manner; its spatial map is validated at edge, active-interior and basal-arc positions (maximum error 1.69% of map amplitude). The covariance-weighted map predicts domain drift without fitting stochastic data.
- Wave-pinning f=0.001 C60/C20 drift remains 0.490 predicted and 0.473 measured; Goryachev remains quantitatively inconsistent. The weak-input alpha bridge retains unavailable observed exponents explicitly.
- Linear range at the right edge (first tested >10% gain departure, with last linear amplitude one doubling below): Wave-pinning: first >10% departures 0.02048 at C=20 and 0.01024 at C=60, size fold 0.5; last tested linear amplitudes 0.01024 and 0.00512; Holmes: first >10% departures 0.0512 at C=20 and 0.0512 at C=60, size fold 1; last tested linear amplitudes 0.0256 and 0.0256; Spring: first >10% departures 0.1024 at C=1 and 0.1024 at C=3, size fold 1; last tested linear amplitudes 0.0512 and 0.0512; Goryachev: first >10% departures 0.32768 at C=20 and 1.31072 at

Figure 3. Size-dependent susceptibility links deterministic perturbations to weak-input domain drift · continued

C=60, size fold 4; last tested linear amplitudes 0.16384 and 0.65536.

Proposed measurements and experimental follow-up

Map terminal domain shift per dose with local weak pulses at edge, interior and basal positions.

Measure both boundaries, centroid and width; compare deterministic susceptibility and covariance-based drift at matched sizes.

Scope, limitations and pending validation

Goryachev absolute drift prediction fails despite small deterministic-map validation error.

Existing weak-input alpha estimates are unavailable for Wave-pinning and Goryachev under the fixed shared-lag rule.

Linear-limit amplitudes are geometric-grid brackets; separated-cue plateau is an operational finite-data median.

Deterministic drift, total heading decay and empirical migration persistence remain distinct.

Model definitions and readouts

$$G(x_0) = \sum_j g_j h(x_j - x_0) \tag{3.1}$$

The main map is the standard Gaussian-pulse response per dose, not raw cell-input g.

$$c_j = g_j \sum_k g_k \int_0^\infty \overline{K_u(j,k;t,t+v)}^{\,t}\,\mathrm{d}v, \qquad D_s = \sum_j c_j \tag{3.2}$$

Signed per-position contributions retain the exact native input covariance.

$$\alpha_{\mathrm{pred}} = 2 - \frac{\mathrm{d}\ln D_s}{\mathrm{d}\ln \bar{C}} \tag{3.3}$$

Same fixed adjacent-size neighborhoods as Figure 4; missing observed alpha remains unavailable.

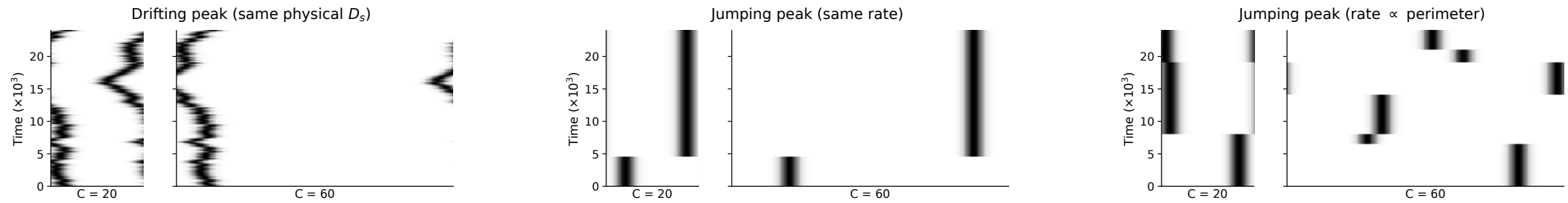
Sources and reproducibility

Scientific methods and unexpected results

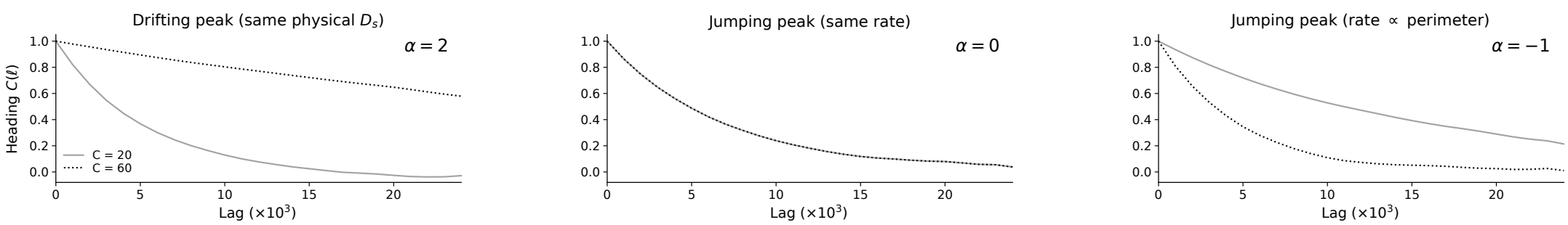
Figure 3. Size-dependent susceptibility links deterministic perturbations to weak-input domain drift · continued

Susceptibility maps and signed contributions
Every deterministic assay
Alpha bridge and unavailable rows

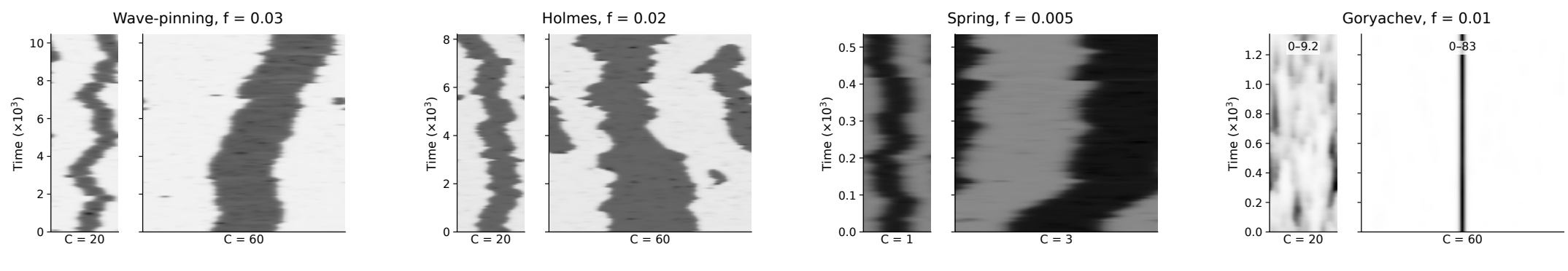
A


Drifting peak (same physical D_s)
Jumping peak (same rate)
Jumping peak (rate ∝ perimeter)
Time (×10³)
C = 20
C = 60

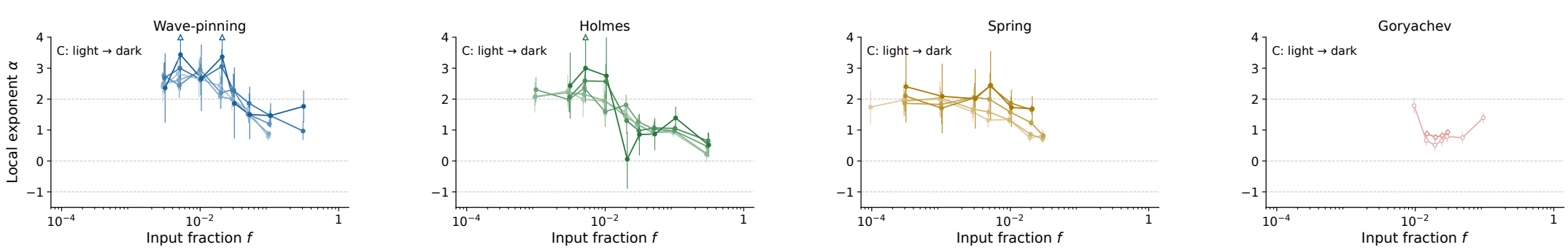
B


Drifting peak (same physical D_s)
Jumping peak (same rate)
Jumping peak (rate ∝ perimeter)
$\alpha = 2$
$\alpha = 0$
$\alpha = -1$
Heading $C(\ell)$
Lag (×10³)
C = 20
C = 60

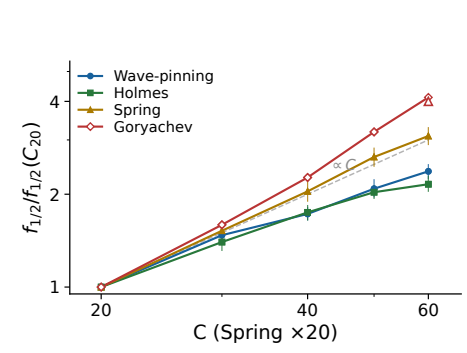
C


Wave-pinning, f = 0.03
Holmes, f = 0.02
Spring, f = 0.005
Goryachev, f = 0.01
Time (×10³)
C = 20
C = 60
C = 1
C = 3
0–9.2
0–83

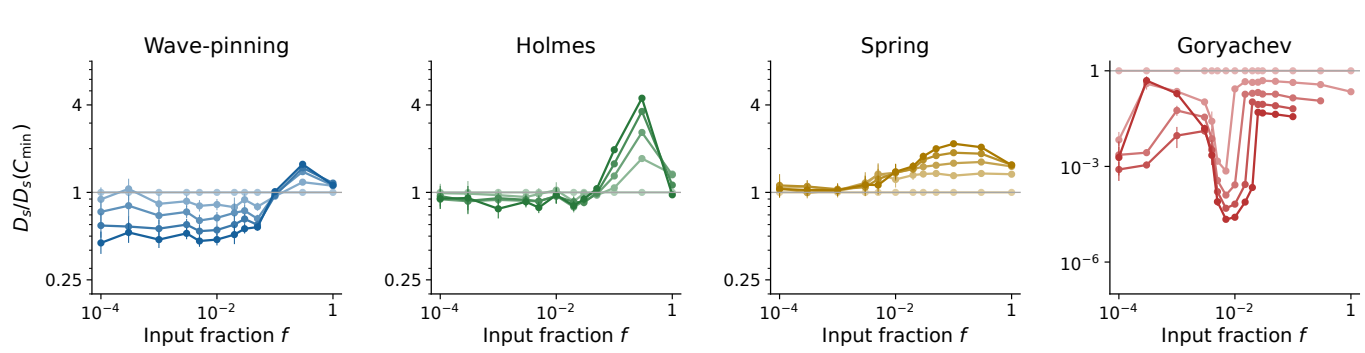
D


Wave-pinning
Holmes
Spring
Goryachev
C: light → dark
Local exponent α
Input fraction f

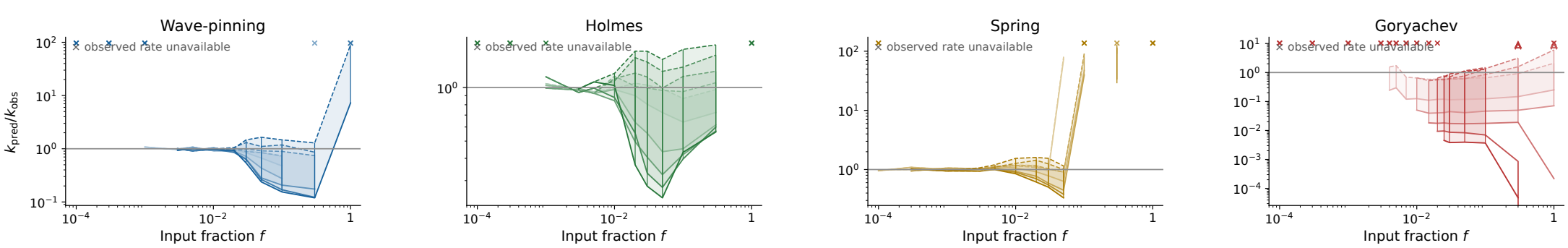
E


Wave-pinning
Holmes
Spring
Goryachev
$f_{1/2}/f_{1/2}(C_{20})$
C (Spring ×20)

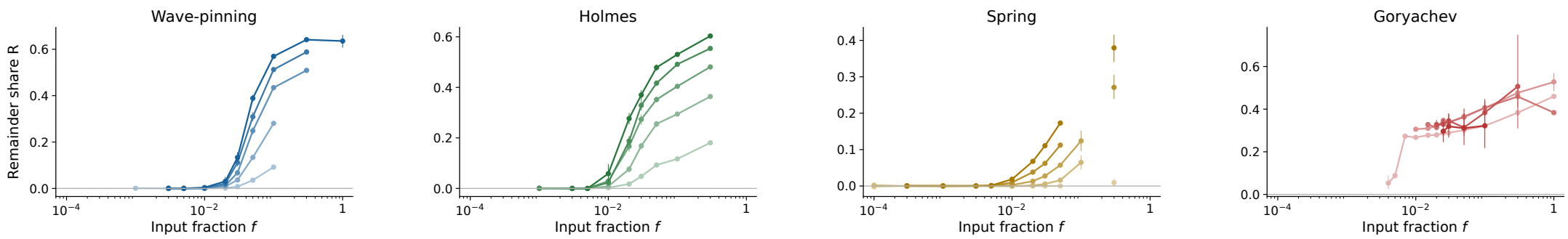
F


Wave-pinning
Holmes
Spring
Goryachev
$D_s/D_s(C_{min})$
Input fraction f

G


Wave-pinning
Holmes
Spring
Goryachev
× observed rate unavailable
k_{pred}/k_{obs}
Input fraction f

H


Wave-pinning
Holmes
Spring
Goryachev
Remainder share R
Input fraction f

I

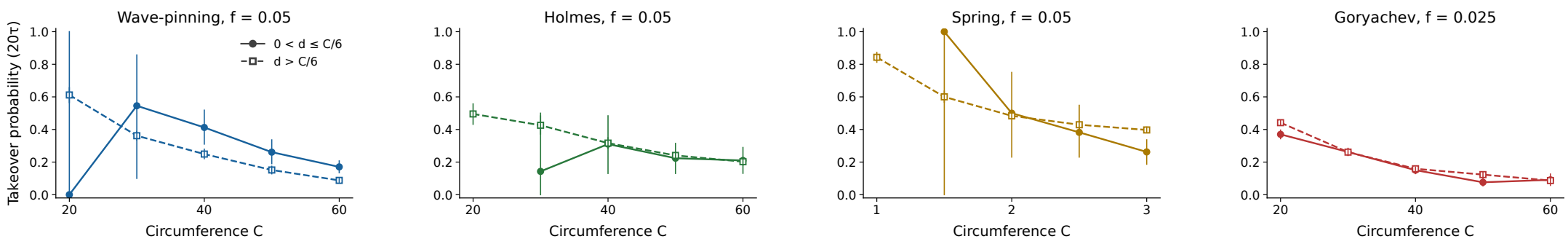

Wave-pinning, f = 0.05
Holmes, f = 0.05
Spring, f = 0.05
Goryachev, f = 0.025
0 < d ≤ C/6
d > C/6
Takeover probability (20τ)
Circumference C

Figure 4. Enlargement preserves heading as noisy polarity changes regime.

Size-dependent heading persistence, region-based diagnostic limits and a detector-free mode test.

Detailed legend

Model panels use unchanged positive lognormal input with mean f times the native reference: Wave-pinning 0.1, Holmes 1, Spring 4 and Goryachev 6.4. The Gaussian driving field has unit variance, lognormal noise amplitude σ = 1, spatial parameter 1.5 and OU correlation time 50 (Spring basal-physical 0.15 and 5). Spring input is stationary in material coordinate, so its physical correlation length scales with circumference. Native clocks are not calibrated between models. Existing statistics comprise 260 conditions and 45,696 histories; targeted independent mode-only histories are additional and do not alter A–I.

A, Detector-free null kymographs at C20 and C60: identical physical diffusion, size-independent jumping, or perimeter-proportional jumping. Each facet shows the same activity measure.

B, Null heading autocorrelation; the corresponding persistence size exponents are 2, 0 and −1. C20 is light and C60 dark.

C, Native-model kymographs at the unchanged round 5 transition inputs: Wave-pinning 0.03, Holmes 0.02, Spring 0.005 and Goryachev 0.01. Positions are centred at the initial heading; time is elapsed after burn-in. Goryachev uses the explicitly labelled per-size activity ranges.

D, Local persistence exponent α versus input fraction f, with the inherited fixed three-size neighbourhood (two size values at endpoints), whole-history 95% intervals and light-to-dark size ramps. In D and F–H, sizes are C = 20, 30, 40, 50 and 60 (Spring 1, 1.5, 2, 2.5 and 3), from light to dark. Open circles retain forced-condition flags; missing rates are unavailable, not zero.

E, Noise-tolerance fold at fixed heading-correlation lag 2000 (Spring 200), relative to the smallest ring. Spring circumference is multiplied by 20 for plotting only.

F, Physical within-tenure drift coefficient normalized by the smallest size at the same input. Native equations and the five-frame dominant-tenure rule are unchanged.

G, Drift plus counted five-frame-dominant takeover closure over every designed input. Solid lower and dashed upper limits include every unscored takeover angle with decorrelation loss between 0 and 2; shaded intervals are partial-identification bounds, not confidence intervals. Grey line: unity. Open upward triangles with arrows mark an unbounded upper limit where drift is unavailable; drift is bounded below by zero there. Crosses at the top mark unavailable observed heading rates. No mean angle is imputed to unscored events.

Figure 4. Enlargement preserves heading as noisy polarity changes regime. · continued

H, Signed remainder share versus input at each size, using the round 5 native-amplitude point-region convention. It contains amplitude, within-region shape, unresolved activity and background; it is descriptive. Exact finite-width moments can be attenuated again by the original native-amplitude scaling, so their nonshrinking remainder does not establish failure of a domain mechanism. Exact own-mask checks appear in the supplement.

I, Certified takeover probability within 20 input correlation times, separately selected per model. Positive birth-to-incumbent distances only; filled circles $0 < d \leq C/6$, open squares $d > C/6$. Wave-pinning f=0.05, minimum total births=590; Holmes f=0.05, minimum total births=231; Spring f=0.05, minimum total births=580; Goryachev f=0.025, minimum total births=1064. The selected designed input is nearest in log distance to the smallest ring's f½ among those with at least 200 total certified births at every size. Intervals resample independent histories 2000 times; [0,1] bounds replace intervals when fewer than 97.5% of resamples have a defined probability. The two distance bins can have very different denominators. Certification and finite follow-up limit inference about all births.

The strong-input regime is currently unexplained. Candidate J–K, spectra and exact-mask diagnostics remain supplementary; two exemplar seeds alone cannot validate collapse. Independent mode-collapse test not yet assessed: targeted precision evidence is incomplete; the strong regime is currently unexplained.

Narrative points and conclusions

- Larger rings preserve heading longer while their local size exponent weakens as input strengthens. Completed region diagnostics show all-input closure bounds and remainder curves. The independent modes-only campaign is active; this review candidate claims no completed mode-collapse result.

Proposed measurements and experimental follow-up

Heading persistence and local size exponent, not migration persistence.

Takeover probabilities are per-model, finite-horizon certified-subset quantities.

Existing exemplar Fourier spectra and exact own-mask diagnostic; targeted mode-only independent histories retain unchanged native equations.

Scope, limitations and pending validation

Missing takeover angles are bounds, not imputed means.

Exact birth/split population classification remains partly unidentifiable.

Two exemplar seeds cannot validate universal mode collapse.

Figure 4. Enlargement preserves heading as noisy polarity changes regime. · continued

Strong-input mechanism remains unexplained.

Immutable review snapshot: the targeted modes campaign and precision additions were active at capture; its final scientific assessment is pending.

The targeted modes-only campaign is active in this immutable review snapshot; final precision results will be integrated after it finishes.

Model definitions and readouts

$$Q_t = \frac{\sum_{i \in I_t} A_{i,t} e^{i\theta_{i,t}}}{\sum_{i \in I_t} A_{i,t}}, \qquad r_t = |P_t|, \quad \widetilde{Q}_t = r_t Q_t \tag{4.1}$$

Q is the normalized point-region moment; P=M₁/M₀ is measured from the native field. Its amplitude r prevents the unit-modulus single-region normalization artefact.

$$\Delta P_j = \bar{r}\,\Delta Q_j \ (j = \mathrm{M}, \mathrm{A}, \mathrm{B}), \qquad \Delta P_{\mathrm{R}} = \Delta P - \bar{r}\,\Delta Q \tag{4.2}$$

The three ΔQ components use the symmetric Shapley convention for motion, coexisting amount and membership; r̄ is mean endpoint native amplitude. R closes the native moment change and includes amplitude variation, shape and background. This is an accounting convention, not causal attribution.

$$L_j = \frac{\mathrm{Re}\left[\Delta P_j \overline{(H_1 - H_0)}\right]}{|P_0| + |P_1|}, \quad H_t = \frac{P_t}{|P_t|}, \quad \sum_j L_j = 1 - \cos(\theta_1 - \theta_0) \tag{4.3}$$

Pooled signed fractions are sums of L_j divided by total measured heading loss. They retain cancellation, can exceed one, and depend on the stated accounting convention; they are not causal shares.

$$k(\ell) = -\frac{\ln C(\ell)}{\ell}, \qquad k \propto \bar{C}^{-\alpha} \tag{4.4}$$

C(ℓ) is activity-heading autocorrelation; mean C is actual circumference. Endpoint eligibility uses |P|≥0.1. Main G evaluates every designed input; unavailable observed rates remain explicit.

$$a_n(t) = \frac{1}{N} \sum_{j=1}^{N} u_j(t) e^{in\theta_j}, \qquad P(t) = \frac{a_1(t)}{a_0(t)}, \qquad q_n = \frac{2\pi n}{\bar{C}} \tag{4.5}$$

Fourier modes use cell-centre material phases and the positive convention matching the native first moment. Heading is the phase of P; Spring physical wavenumber uses mean circumference.

Figure 4. Enlargement preserves heading as noisy polarity changes regime. · continued

$$\gamma_n = -\frac{\ln C_n(\ell_*)}{\ell_*}, \qquad k(C) = \gamma(2\pi/C), \qquad \alpha = \frac{\mathrm{d}\ln\gamma}{\mathrm{d}\ln q} \tag{4.6}$$

The last two equalities are the detector-free hypothesis under a common size-independent mode curve, not an identity assumed in analysis. Complex correlations use the nearest-half lag rule; omitted half crossings are explicit bounds. Independent mode-collapse test not yet assessed: targeted precision evidence is incomplete; the strong regime is currently unexplained.

$$k^{\mathrm{lo}}_{\mathrm{pred}} = q_1^2 D_s + \frac{\sum_{\mathrm{scored}} (1-\cos\Delta\theta)}{T}, \qquad k^{\mathrm{hi}}_{\mathrm{pred}} = k^{\mathrm{lo}}_{\mathrm{pred}} + \frac{2N_{\mathrm{unscored}}}{T} \tag{4.7}$$

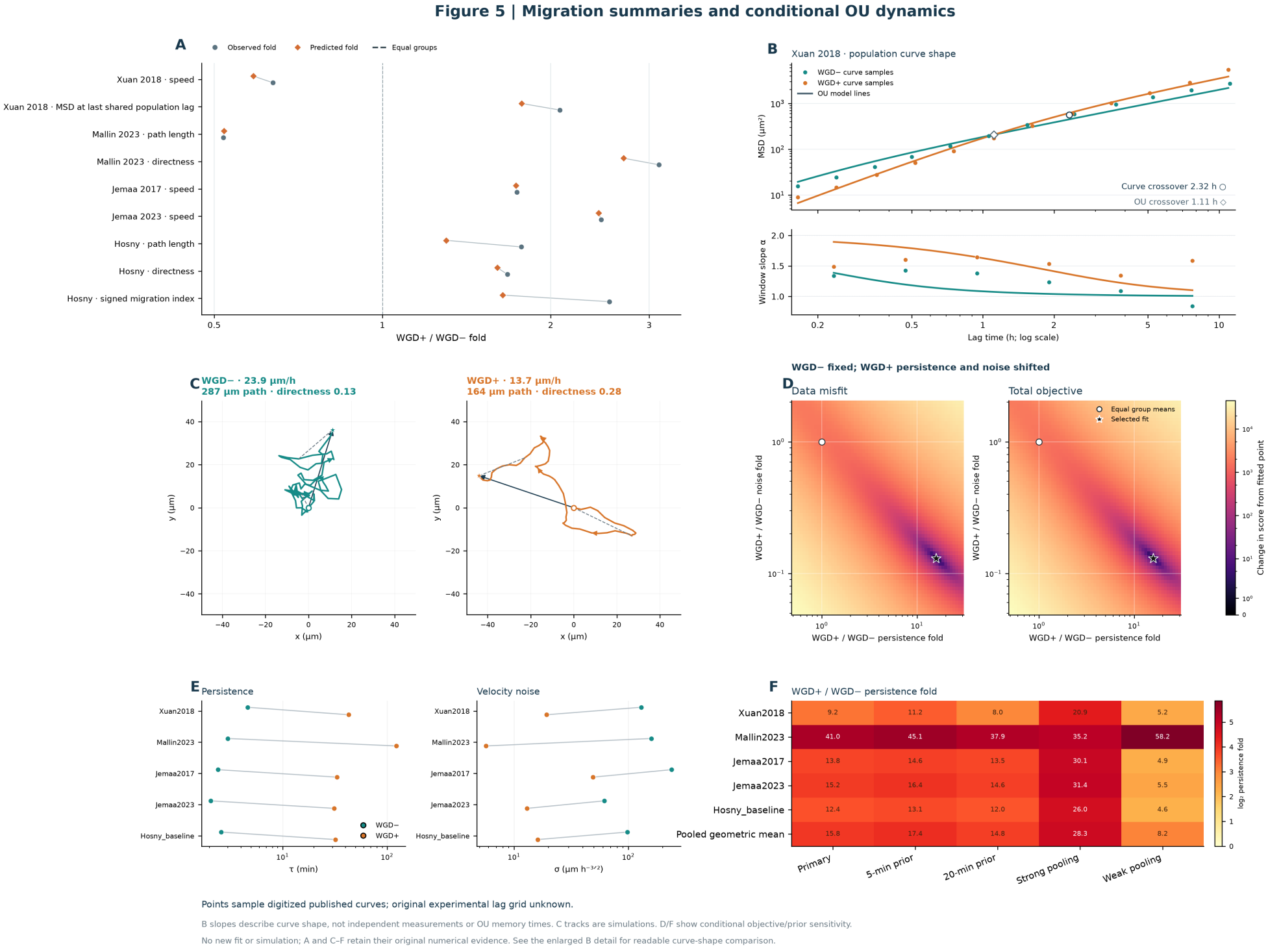
All counted takeovers enter the interval: each unscored angular loss lies between0 and2. T is inherited eligible exposure. Missing drift has a nonnegative lower bound and unbounded upper limit; an unavailable observed heading rate prevents a numerical ratio.

Sources and reproducibility

Snapshot methods

Snapshot results and active-campaign status

Figure 5. Migration observations and conditional OU interpretation · image 1 of 2



Reassembled Figure 5: original A and C–F numerical evidence, revised MSD/curve-shape panel B.

Figure 5. Migration observations and conditional OU interpretation · image 2 of 2

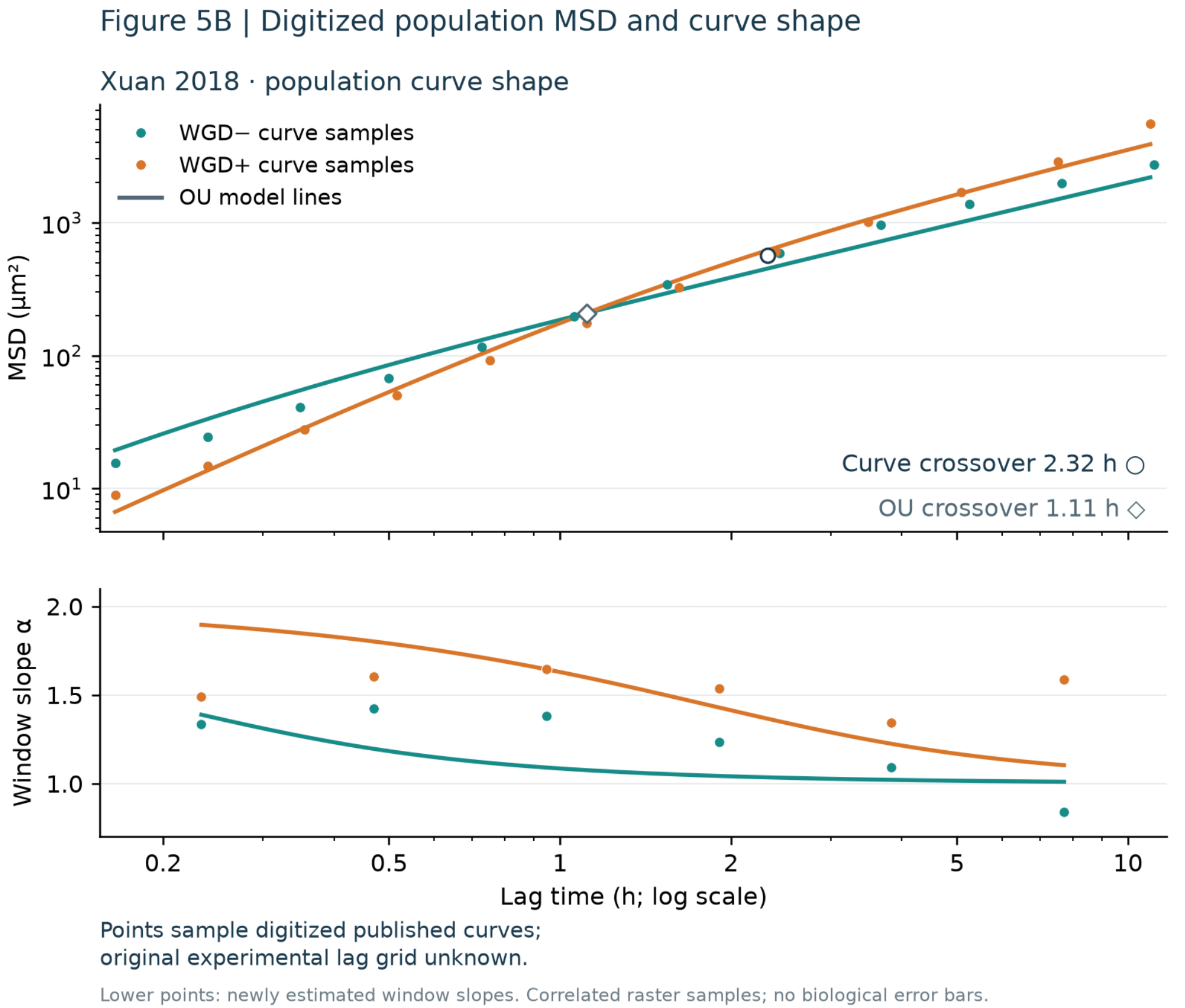


Figure 5B continued: enlarged detail with explicit raster-sample caveat.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation

Existing empirical support, with observed migration summaries kept separate from fitted velocity-memory parameters.

Detailed legend

A, Published/digitized enlarged or polyploid-group to reference-group ratios for nine migration summaries, alongside primary Ornstein–Uhlenbeck (OU) fit predictions. Xuan contributes sampled mean speed and population mean-squared displacement (MSD) at the last shared lag, approximately 10.95 h; Hosny summaries are group medians. Mallin central-bar statistic is not specified by the source and retains the historical mean approximation. These distinct source summaries are not interchangeable measures of persistence.

B, Upper: twelve chosen display samples from each digitized Xuan population MSD curve (small points) and the unchanged fitted analytical OU expectation (solid lines), on logarithmic lag and MSD axes. These are samples of the published raster curves, not recovered experimental lag positions or independent cell observations; the original experimental lag grid is unknown. The open circle identifies the crossover of the full digitized traces at approximately 2.32 h, computed by linear interpolation in native lag and MSD. The open diamond identifies the different fitted OU crossover at approximately 1.11 h. Lower: descriptive slopes of log(MSD) versus log(lag) fitted in six equal log-lag windows of the full digitized curves (points), and matching-window moving OU slopes (solid lines). Each window spans approximately 0.304 decades; both observed and model slopes use the same full window support. Observed raster columns remain correlated and give no biological sample size or confidence interval. These slopes are newly calculated curve-shape summaries, not author-reported local exponents or fitted OU memory times. A constant power-law exponent would appear horizontal; reference values 1 and 2 indicate diffusive and straight-line scaling. The model does not reproduce the complete observed curvature or the observed crossover. The analytical fit, all fitting targets and numerical inputs to A and C–F are unchanged.

C, Representative Xuan primary-fit model-generated tracks, with sampled speed, total path length and directness (net displacement divided by path length). Both groups use identical spatial scales. The tracks are simulations, not published individual-cell trajectories; a slower sampled trajectory can nevertheless have greater endpoint directness. Dashed chords illustrate the contribution of finite-time displacements.

D, With reference-group parameters fixed, vary enlarged-group persistence and velocity-noise ratios. Separate surfaces show change in the descriptive data-misfit score and in the total regularized objective; the selected fit and equal-group reference are marked. The heuristic score is not a calibrated likelihood, and the surfaces are not confidence regions.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

E, Primary paired persistence and velocity-noise estimates for Xuan 2018, Mallin 2023, Jemaà 2017, Jemaà 2023 and the Hosny hypoxic baseline. F, Enlarged/reference persistence folds under the primary 10-minute prior, 5- and 20-minute alternatives and stronger/weaker pooling, with the pooled geometric mean. Priors and pooling define conditional sensitivity scenarios, not uncertainty intervals. The Jemaà comparisons constrain speed alone, leaving persistence/noise tradeoffs dependent on hierarchy and priors.

The fitted population contrasts span stable tetraploids, giant polyploids and treatment-induced polyaneuploid cancer cells (PACCs). The inherited WGD+/WGD− labels in the artwork are comparison shorthand and do not establish matched WGD-only interventions in every study. Hosny observations derive from the recovered preprint; equivalence to the journal version remains unresolved.

Narrative points and conclusions

- Xuan combines lower sampled speed (0.64-fold) with greater late-lag population MSD (2.08-fold); Mallin combines shorter path length (0.52-fold) with greater directness (3.12-fold). Persistence-related behaviour therefore deserves analysis separately from speed.
- Hosny baseline PACCs have greater median path length (1.77-fold) and directness (1.67-fold). Signed migration index is a separate directional summary.
- All five primary OU comparisons infer longer velocity memory. The pooled primary persistence fold is about 15.8, and remains above one across the declared prior/pooling scenarios; this is a conditional model result.
- The fit does not reproduce the complete Xuan curve shape within its raster envelope, and data-only sensitivity has rank 18 of 22. Optimization completion does not establish model adequacy or uniquely identified persistence times.
- The observed Xuan curves cross near 2.32 h while the unchanged fitted OU curves cross near 1.11 h. Matched-window slopes make the remaining curve-shape discrepancy visible; fitted persistence is a conditional summary.
- Lower sampled speed can coexist with greater late-lag displacement in the enlarged-group comparison. Its causal connection to polarity mechanisms requires matched activity and migration measurements.

Proposed measurements and experimental follow-up

The new WGD assay should measure matched trajectories directly, with declared sampling and observation duration, and assess speed, displacement, directness and lag-dependent MSD before considering fitted persistence. An activity-polarity memory

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

measurement would complement these quantities without equating activity and velocity.

Scope, limitations and pending validation

Neither these heterogeneous group comparisons nor the OU fits establish that enlargement alone causes persistence. Cell identity, induction/treatment and acquisition differences require explicit treatment in the new experiments.

The requested observed-point/model-line and curve-shape display is implemented. The original experimental lag grid remains unknown; neither display samples nor raster-derived slopes supply independent biological replicates. The observed/model crossover discrepancy remains visible.

Sources and reproducibility

Refined OU analysis
Original panel legend
Observed folds
Fit results and adequacy
Hosny sensitivity supplement
Revised display, exact preserved inputs and direct-user feedback
Migration display numerical and preservation checks

Figure 6. RESERVED: new WGD motility results

RESERVED MAJOR EXPERIMENTAL RESULTS

New WGD motility assays — all data pending

A Paired biological system

SUM-159 chemically induced WGD+/− pilot

Confirm group identity, size and morphology

PENDING — no observations available

B Imaging and observed tracks

Optimize acquisition and tracking

Declare cadence, duration, eligibility and replicates

PENDING — no observations available

C Measured migration behaviour

Speed, path length, displacement and directness

Lag-dependent MSD under declared observation rules

PENDING — no observations available

D Evaluate the persistence hypothesis

Relate the acquired observations to WGD and size

Retain null or opposite outcomes; specify controls

PENDING — no observations available

Proposed panel plan only. Design and analysis remain to be agreed before interpreting data.

Explicit placeholder. Every panel is pending acquisition, analysis and author review; no mock observations are displayed.

Figure 6. RESERVED: new WGD motility-assay results

A substantial new experimental results section and main figure. No observations are available; the panel layout is a proposal, not an experimental result.

Detailed legend

A, Proposed experimental context: the SUM-159 chemically induced WGD+/− pair is the agreed starting pilot with the Speros collaboration. Imaging should first be optimized for that pair before expansion to additional cell lines. Ploidy/group identity, physical size and morphology need to be documented so the intended comparison is clear.

B, Reserved for representative experimental fields and trajectories under the selected acquisition conditions. The eventual panel should state sampling interval, usable duration, spatial scale, track eligibility and exclusions. Number of independent preparations and biological replicates remains to be specified.

C, Reserved for empirical motility summaries: sampled speed, accumulated path length, endpoint displacement/directness and lag-dependent MSD. The final readouts, follow-up window and analysis plan are to be settled before interpreting the data; this placeholder does not imply that all listed measurements will be technically feasible.

D, Reserved for the measured relationship between WGD-associated enlargement and migration behaviour, including size distributions and controls needed to interpret the group contrast. Any activity/protrusion measurements arising from the mechanism proposals require a separately agreed assay and should be distinguished from the basic motility pilot. Statistical models, uncertainty reporting and sample-size targets remain open.

Narrative points and conclusions

- This major results section will test whether the planned paired experimental system exhibits the migration behaviour motivating the theoretical work.
- Its conclusions must follow the acquired observations, including null or opposing results. Simulation output and literature fits cannot fill this gap.
- Agreement between increased polarity resistance or activity memory and migration persistence would support the proposed connection; causal attribution requires appropriate perturbations and controls.

Proposed measurements and experimental follow-up

Pending decisions include plating density, acquisition duration and cadence, field/condition throughput, tracking and exclusion rules, replicate structure, confirmation of WGD status, size quantification and the statistical analysis. The known throughput versus spatial/temporal resolution trade-off should be resolved by the pilot.

Figure 6. RESERVED: new WGD motility-assay results · continued

Mechanism validation can subsequently examine incumbent extent, early challenger growth, local loss/rebound, site fates and global activity-polarity correlation under characterized perturbations. These activity experiments are proposals alongside the theoretical mechanisms and are separate from the reserved motility-results contribution.

Scope, limitations and pending validation

No pilot acquisition, sample size, effect estimate, significance claim or experimental outcome is supplied by the available evidence.

Sources and reproducibility

Author manuscript vision and supplied collaboration context

Supplementary material and supporting analyses

Figure 1: round6 complete supporting evidence

Current source data, numerical qualification and preceding supplementary evidence; earlier main artwork is preserved.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round6_20261023/figure1/figure1_round6_supplement.pdf

Scientific results and rebuild instructions

Figure 2: round6 complete supporting evidence

Current source data, numerical qualification and preceding supplementary evidence; earlier main artwork is preserved.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round6_20261023/figure2/figure2_round6_supplement.pdf

Scientific results and rebuild instructions

Figure 3: round6 complete supporting evidence

Current source data, numerical qualification and preceding supplementary evidence; earlier main artwork is preserved.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round6_20261023/figure3/figure3_round6_supplement.pdf

Scientific results and rebuild instructions

Figure 4: round6 complete supporting evidence

Current source data, numerical qualification and preceding supplementary evidence; earlier main artwork is preserved.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round6_20261023/figure4/supporting/review_snapshot_initial/figure4_supplement.pdf

Scientific results and rebuild instructions

Prospective SUM-159 WGD pilot observables

Specified on 23 October 2026, before SUM-159 WGD+/− motility data; no experimental observations were used.

Model drift subset: f ≤ 0.01 and multiple-domain fraction ≤ 0.05. In cells, classify by the observed frequency of coexisting protrusions; model f and native times have no calibrated cellular equivalents.

Cellular observable	Model counterpart	Drift regime: WGD+ / WGD−	Secondary-domain regime: WGD+ / WGD−	Mechanistic interpretation
Common-mode motion of the leading region's two lateral edges	Correlated boundary translation and domain drift (Figs 3D, 4D)	Similar (Holmes/Spring); smaller in Wave-pinning/Goryachev	No common direction identified outside persistent single-domain bouts	Translation; does not identify a Figure 2 route
Secondary-protrusion birth rate per unit perimeter	Persistent no-overlap birth rate (Fig. 4E(i))	Rare in both groups	Model dependent: Spring higher and Goryachev lower at f = 0.1; other trends unresolved	Initiation; model-dependent size response
Takeover incidence within a fixed follow-up window	Five-frame dominance within 20τ after birth (Fig. 4E(ii))	Not applicable when births are too rare	Lower by 20τ; persists through 80τ in the same cohort; lifetime fate unidentified	Incumbent resistance; lifetime success unresolved
Activity/protrusion-heading autocorrelation time	Heading decay and size exponent (Fig. 4B–D)	Longer; approximately size squared only when physical drift is size independent	Size advantage shrinks in the tested models; compare coherence and domain counts	Regime readout; no unique route assignment
Edge displacement after boundary-straddling photoactivation	Weak deterministic edge response (Fig. 3C)	Smaller (Wave-pinning/Goryachev); similar (Holmes/Spring)	Not specified: the pulse assay uses a stable polarized incumbent	Susceptibility within a route, not a unique mechanism

Prospective directions, not SUM-159 observations or a full statistical preregistration. Freeze sampling, segmentation, physical pulse dose and group inference before comparison; model f and native times have no calibrated cellular equivalents.

Birth rates are model dependent: at f = 0.1, bounds identify a Spring increase and a Goryachev decrease with size. Takeover within 20τ is fixed-horizon incidence; its decline persists through 80τ in the same cohort, while lifetime fate and certification selection remain unresolved.

Common translation tests the drift mechanism. Edge pulses refine susceptibility within routes: Wave-pinning and Holmes share an expansion-constrained class in Figure 2 but predict different edge responses. Combine these readouts with Figure 2 first-rise, width and pulse-gap assays to test initiation-limited reset, saturation/expansion and retained progress after visible activity loss. Activity heading, observed migration and OU-fitted memory remain separate readouts.

Prospective pilot observables before SUM-159 data

Specified on 23 October 2026, before SUM-159 WGD+/− motility data; no experimental observations were used.

Detailed legend

Prospective directions, not SUM-159 observations or a full statistical preregistration. Freeze sampling, segmentation, physical pulse dose and group inference before comparison; model f and native times have no calibrated cellular equivalents.

Birth rates are model dependent: at f = 0.1, bounds identify a Spring increase and a Goryachev decrease with size. Takeover within 20τ is fixed-horizon incidence; its decline persists through 80τ in the same cohort, while lifetime fate and certification selection remain unresolved.

Common translation tests the drift mechanism. Edge pulses refine susceptibility within routes: Wave-pinning and Holmes share an expansion-constrained class in Figure 2 but predict different edge responses. Combine these readouts with Figure 2 first-rise, width and pulse-gap assays to test initiation-limited reset, saturation/expansion and retained progress after visible activity loss. Activity heading, observed migration and OU-fitted memory remain separate readouts.

Narrative points and conclusions

- Model drift subset: f ≤ 0.01 and multiple-domain fraction ≤ 0.05. In cells, classify by the observed frequency of coexisting protrusions; model f and native times have no calibrated cellular equivalents.

Proposed measurements and experimental follow-up

Track both lateral half-contrast boundaries of a persistent leading activity/protrusive region. Report their mean-squared physical displacements and cross-correlation over declared lags, including failed single-domain eligibility and changes in region identity. Compare physical displacement before normalizing to perimeter. The two edges are strongly correlated in model exemplars, so report their covariance rather than assuming independence. Report common-mode displacement (left plus right)/2 separately from width fluctuations (right minus left); a single-edge MSD alone cannot distinguish these modes.

With full images retained, call a new connected protrusive/activity region a birth only if it overlaps no region in the preceding frame and persists for at least five consecutive saved frames. Divide by observed perimeter-time exposure; count splits separately. Freeze segmentation, cadence and overlap rules before group comparison.

For each qualifying birth, measure shortest centroid arc distance from the incumbent minus the incumbent half-width, estimated as excess amount divided by twice the declared plateau height. A takeover requires five consecutive frames of dominance.

Prospective pilot observables before SUM-159 data · continued

Report right censoring and a common follow-up horizon; distinguish all births from any subset whose identity is certifiable. The revised model comparison uses a common 20-noise-correlation-time follow-up and pools $f > 0.01$; these conditioning choices must be separated from the cellular analysis. This is fixed-horizon incidence, not a lifetime takeover probability. Record time since birth, cumulative incidence and right censoring. In the same model cohort with at least 80τ follow-up, the decline persists from 20τ to 80τ; finite observation and certification selection still prevent a population lifetime claim.

Define the activity/protrusion vector from the full spatial signal; estimate heading autocorrelation at declared lags with a fixed coherence rule and report excluded pairs. Compare groups at matched analysis lags and observation duration. Keep cell displacement/velocity persistence as a separate observed readout.

Apply a weak, brief photoactivation spot centred on a lateral leading-edge boundary, matched in physical width and duration across ploidy groups. Track both edges through relaxation, subtract a no-pulse control, and check half/double amplitudes. A response supports a mechanism only to the extent that its size trend differs among model predictions. At model $f = 0.001$, largest/smallest pulse-response gains are 0.701 (Wave-pinning), 0.914 (Holmes), 0.989 (Spring), and 0.0878 (Goryachev); these are native-model size contrasts, not numerical cellular fold predictions. Separately repeat the Figure 3A–B assay with cue placements straddling and separated from the incumbent boundary, measuring half-turn time to distinguish sliding from takeover; the weak local pulse and the finite turning cue are distinct assays.

Scope, limitations and pending validation

No SUM-159 motility data are available in this revision. Directions are prospective and must not be described as confirmed experiments.

Native model times and f are not calibrated to cells. Event holds measured in frames depend on acquisition cadence; select and report that cadence before comparing experimental groups.

The compact model records do not identify every birth, split or local plateau height. Model bounds and certified-subset probabilities do not establish the unconditional cellular birth/takeover directions.

The new observables do not uniquely distinguish the Figure 2 response classes. Weak boundary susceptibility and finite-cue sliding/takeover are different properties; the within-20τ size contrast cannot distinguish slower completion from a change in eventual success.

Sources and reproducibility

Prospective directions and operational definitions

Supplementary material and supporting analyses

Figure 3: edge-response / secondary-domain revision supplement

New measurements, identification limits, numerical checks and preceding supplementary evidence.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round5_20261023/figure3/figure3_round5_review_supplement.pdf

Physical diffusion prediction and observed drift

Size-normalized prediction and observed drift

Normalized integrated boundary response

Methods and covariance derivation

Both boundary assays and numerical checks

Reviewed main/supplement preservation

Figure 4: edge-response / secondary-domain revision supplement

New measurements, identification limits, numerical checks and preceding supplementary evidence.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round5_20261023/figure4/figure4_supplement.pdf

Round5 complete campaign analysis

Scientific identity/enclosure validation

Birth and heading-loss conditions

Takeovers with common follow-up

Original campaign manifest

Native-amplitude heading accounting

Native-amplitude condition metrics

Figure 4: corrected event scoring and full closure comparisons

Scorable-event coverage, missing-angle bounds, both detector-free cadences and all closure failures. Preceding editions remain linked below.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round4_20261023/figure4/figure4_supplement.pdf

Scientific methods and regime-specific counts

Figure 4: round-3 analysis and null comparisons

Pointwise and smooth size exponents, drift/jump cross-checks, sensitivity, valid fractions, replication and analytic null checks. Missing estimates and finite-rate precision shortfalls are identified in the scientific readme.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure4/figure4_supplement.pdf

Unexpected results, data status and methods

Supplementary material and supporting analyses · continued

Figure 2: round-3 evidence and preserved prior panels

The complete supplement adds the per-width replacement curves and the deBelly native endpoint audit, then preserves the complete round-2 main and every earlier supplement page. Superseded fate maps, ratios and class tables are historical evidence.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure2/figure2_round3_supplement.pdf

Round-3 reader and source data

Exact preservation receipt

Figure 3: round-3 evidence and every preserved round-2 panel

The supplement retains 90/99%-heading time curves, the preceding four-placement round-3 main and legend including C = 47.6 profiles, and the complete round-2 main and supplement. Original source files remain unchanged.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure3_single_C/figure3_round3_supplement.pdf

Round-3 reader and native source coordinates

Exact supplement preservation

Historical Figure 4: previous supplement and decision record

Historical evidence from the preceding Figure 4 edition, preserved unchanged. The comprehensive supplement retains the full designed condition grid, unavailable measurements, finite-horizon bounds, native histories, controls and the stated numerical and observational sensitivities. Earlier figure versions remain development history. S0–S12 statistical core and complete native-atlas appendix index; native pages are in all linked appendix volumes. Every numbered appendix below is part of the complete supplement.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/signature_figure4_20261020/supporting/vectors/figure4_revision_complete_supplement.pdf

Figure reader and source data

Independent completion review

S11 native atlas appendix 001

S11 native atlas appendix 002

S11 native atlas appendix 003

S11 native atlas appendix 004

S11 native atlas appendix 005

S11 native atlas appendix 006

S11 native atlas appendix 007

S11 native atlas appendix 008

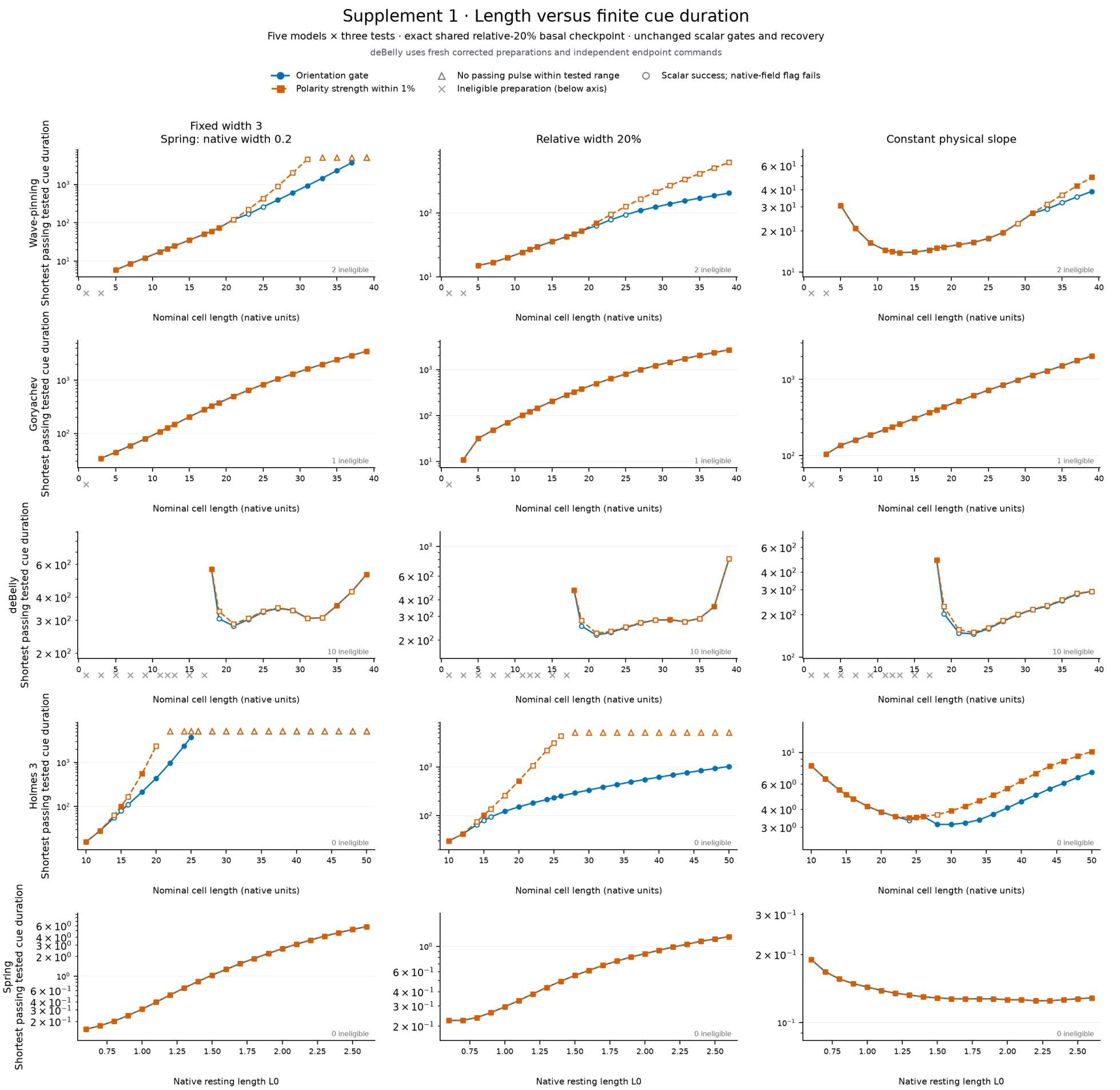
S11 native atlas appendix 009

S11 native atlas appendix 010

Supplementary material and supporting analyses · continued

S11 native atlas appendix 011
S11 native atlas appendix 012
S11 native atlas appendix 013

S1. Full cue-duration response curves



Both patches use a constant peak across all lengths; Goryachev K=64, peak=6.4. Gradients retain one constant physical slope per model.
Each positive duration restarts the original checkpoint. Full strength adds the 1% reversed initial-polarity band to the orientation gate.
Open markers retain failing field-tail flags. Censoring marks explicit tested limits; missing endpoints break lines. Native times and scales differ.
Data generated 2026-10-08T19:26:33.200718+00:00; producer finished. Every imported and fresh trial passed independent native-array audit.

Canonical corrected five-model duration curves; source working number Supplement 1.

Earlier supporting material: S1. Full cue-duration response curves

Essential curve context for the main comparison heatmap.

Detailed legend

Five model rows and three finite-cue columns retain the complete sampled length dependence. Coloured curves distinguish orientation from strength replacement. Open success markers retain scalar successes that fail the separate native-field tail flag; triangles indicate no passing pulse within the tested range; crosses indicate ineligible preparations. Missing values are not zero response times. Each duration excludes preparation and the subsequent 10,000-unit basal recovery. Spring uses native length and time. See Figure 1 for exact gates and stimulus conventions.

Narrative points and conclusions

- Corrected deBelly is nonmonotone across all three geometries. The positive rank coefficients summarize association rather than its full curve shape.
- Constant-slope Spring shows the opposite association to its patch responses. Holmes orientation under that cue varies with little monotone association. Censoring and native-field flags restrict interpretation even where rank correlation is strong.

Sources and reproducibility

Current figure legend and gate definitions

Supplementary material and supporting analyses

Earlier supporting material: S2. Numerical qualification and preparation provenance

Exact common starting states, duration-boundary audits, corrected deBelly endpoint controls and selected mesh diagnostics; source histories remain distinguishable from compact committed summaries.

polarity1d/manuscript/THREE_TESTS.md

Earlier supporting material: S3. Mechanism and stochastic sensitivity

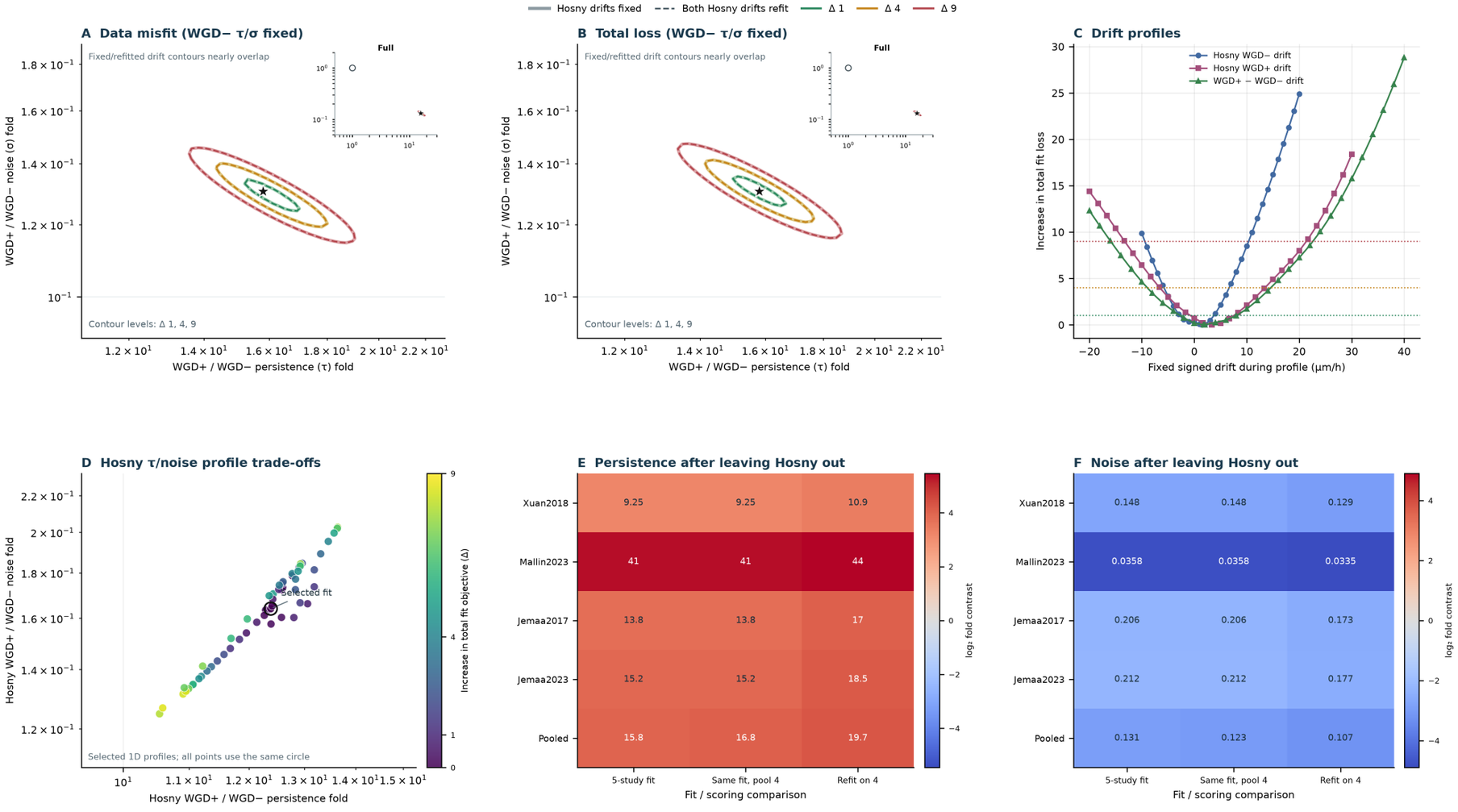
Complete gain/recovery grids, clock qualification, finite-cue definitions, outcome tables, site/vector diagnostics and numerical/noise controls accompany the mechanism checkpoint. The historical Figure 4 edition also displays both strengths, complete half-window diagnostics, physical-site eligibility and matched C20/C60 native examples; compact source tables support portable rerendering.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/VISION_CHECKPOINT_REQUIREMENTS.md

S4. Migration-model adequacy and Hosny sensitivity

How much does the Hosny drift assumption change the fitted contrast?

Primary Gaussian OU hierarchy · profile panels reoptimize remaining parameters · Δ values are descriptive fit sensitivity



Objective is arbitrarily weighted and has no calibrated statistical or biological units; it is not a likelihood. Profile ranges are unions of sampled 1D fits, not joint uncertainty regions. Hosny contributes medians to fitting.

Existing Hosny sensitivity supplement from the refined OU analysis.

Earlier supporting material: S4. Migration-model adequacy and Hosny sensitivity

Supporting conditional sensitivity analysis; it is not an independent biological replication.

Detailed legend

The retained supplement explores signed-drift and leave-Hosny-out sensitivity within the declared OU hierarchy. Its fitted changes and descriptive-score profiles are conditional on the model and supplied summaries, not confidence intervals. Data-only identifiability, Xuan population-curve residuals and prior/pooling assumptions remain part of the adequacy assessment. Representative movie diagnostics are separate: acquisition timing is uncertain and nuclear appearance does not establish ploidy.

Sources and reproducibility

Original Hosny supplement caption

Sensitivity methods and results

Earlier supporting material: Gain and small-loss recovery: supporting evidence

Detailed legend

Calculated native input gain is separate from observed post-cue activity. The small-loss recovery contrast persists through stored finite trajectories but weakens with time; actual damage is only about0.22–0.285% of core activity, with no reporter calibration.

Scope, limitations and pending validation

Small-loss assays are not the finite optical challenge or a universal stability measure.

Sources and reproducibility

Finite-time and amplitude audit

All97 gain/recovery values

Earlier supporting material: Selected native pulse and causal controls

Detailed legend

WP L15 T20/T60 starts from the same basal state and illustrates withdrawal-dependent domain survival. Evolved interventions have endpoint- and cue-specific limits.

Scope, limitations and pending validation

WP4× mobility passes orientation only at one dose; both conditions fail the stronger full-polarity-strength gate. Mobility is changed during cue and recovery.

Sources and reproducibility

Retained WP short/long pulse artwork

Endpoint-qualified controls

Supplementary material and supporting analyses

Earlier supporting material: Stationary-cue endpoint-vector alternative

An alternative view displays all 240 measured endpoint vectors on equal coordinate scales, with weak-resultant, overlap and finite-tail flags retained. It is available for discussion without replacing the complete withdrawal/retention map in the current review.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/standard_angles_sizes/vector_grid/README.md

Earlier supporting material: Spring: every native medoid

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_all_histories.png

Earlier supporting material: Spring magnitude, geometry and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_magnitude_geometry_controls.png

Earlier supporting material: Spring full/half-window memory

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_memory_windows.png

Earlier supporting material: Spring increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_increment_tails.png

Earlier supporting material: Spring short/long-lag geometry and circular mixing

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_endpoint_lag_sensitivity.png

Earlier supporting material: Spring largest sampled increment native context

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_large_increment_context.png

Earlier supporting material: All chemical models including Goryachev

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_stochastic_revision.png

Earlier supporting material: Chemical memory: both strengths

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_both_strengths_memory.png

Supplementary material and supporting analyses · continued

Earlier supporting material: Chemical magnitude and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_magnitude_controls.png

Earlier supporting material: Chemical increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_increment_tails.png

Earlier supporting material: Chemical site-gate sensitivity

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_site_gate_sensitivity.png

Earlier supporting material: Original chemical medoids, all18 conditions

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_selected_histories.png

Earlier supporting material: Goryachev lowest-resultant diagnostic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_lowest_resultant_diagnostic.png

Earlier supporting material: Spring active amount and spatial first harmonic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_amount_spatial_balance.png

Earlier supporting material: Preserved complete mechanism and assay checkpoint

The preceding outline retains all-five mechanism-to-measurement sheets, ring gain/recovery and finite-cue maps, and noise magnitude, controls and full/first/last-window diagnostics. Its numerical evidence is preserved; the revised main-figure claims are stated in this edition.

manuscript_outline/outline_v2.pdf

Prior all-five mechanism evidence crosswalk

Prior complete ring and noise checkpoint

Prior physical motion, magnitude and coverage evidence

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope
Figures 2–4 revision specification
Figure 2 October 20 revision brief
Figure 3 October 20 revision brief
Figure 4 October 20 revision brief